

Deutsche Bank
Research

Tracking AI adoption, spending, and capex (Nov 2026)

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IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. UNTIL 19th MARCH 2021 INCOMPLETE DISCLOSURE INFORMATION MAY HAVE BEEN DISPLAYED, PLEASE SEE APPENDIX 1 FOR FURTHER DETAILS.

1.
Summary



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Summary of recent developments in AI indicators

Category	Indicator	Period	Latest value	Change from previous	Change YoY
Adoption	Firms using AI	Oct-26	24.6%	+0.8ppt	+7.3ppt*
Adoption	Firms expecting AI use	Oct-26	28.6%	+0.9ppt	+7.5ppt*
Adoption	Businesses paying for AI tools	Sep-26	55.9%	+0.7ppt	+12.1ppt
Adoption	GenAI use for work	May-26	43.0%**	N/A	N/A
Consumer	GenAI use for personal/work life	May-26	58.0%***	N/A	N/A
Labor Market	AI job posting share	Sep-26	7.34%	+0.6ppt (MoM)	+3.6ppt
Investment	Data-center construction spending	Sep-26	$96.4bn	+8.6%	+90.3%
National Accounts	Information Processing Equipment Investment	Q3-26	$854.9 bn	+37.3% (% QoQ AR)	+29.1%

Note: *Change YoY between 202524 and 202621; ** Average of Gallup, Bick, Blandin and Deming (BBD) and Hartley et al.(HJMM) surveys ; *** BBD survey datapoint

Source: Census, Ramp, Stanford, Indeed, BEA, Haver Analytics, Deutsche Bank

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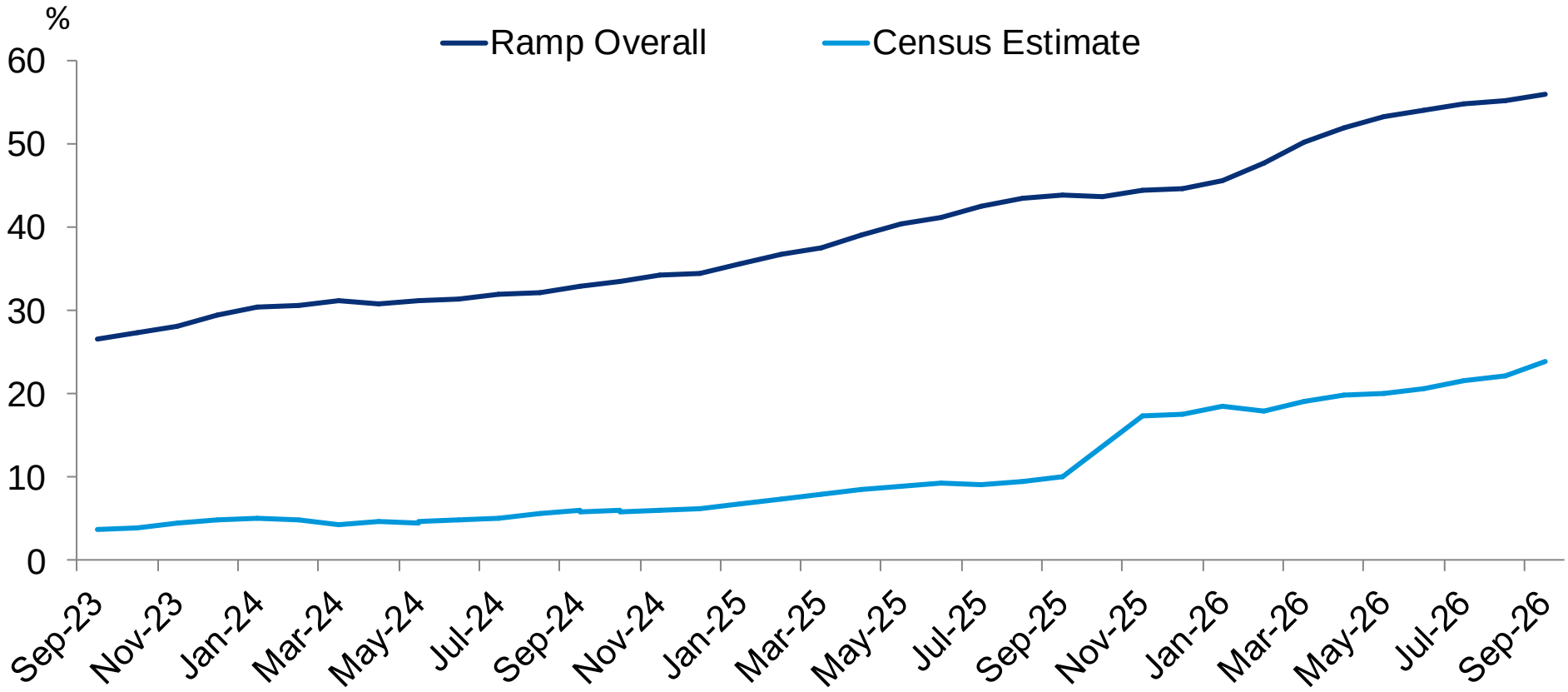
2.
Firm adoption / usage



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AI adoption rates have continued to rise, though alternative datasets disagree on the levels

AI adoption

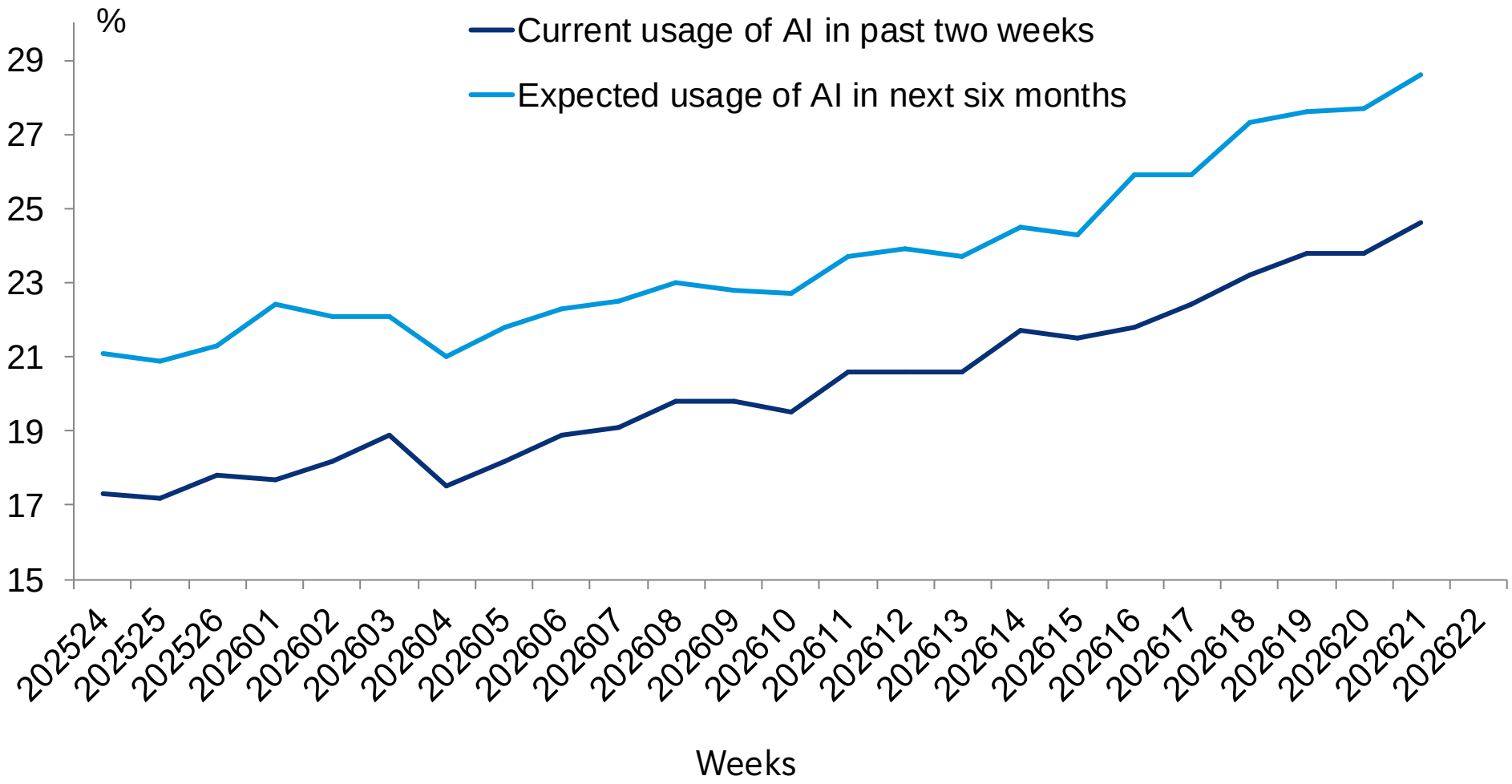


Source: Ramp, Census Bureau, Deutsche Bank

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Census data show expectations for continued expansion in AI adoption over coming six months

% of firms using AI

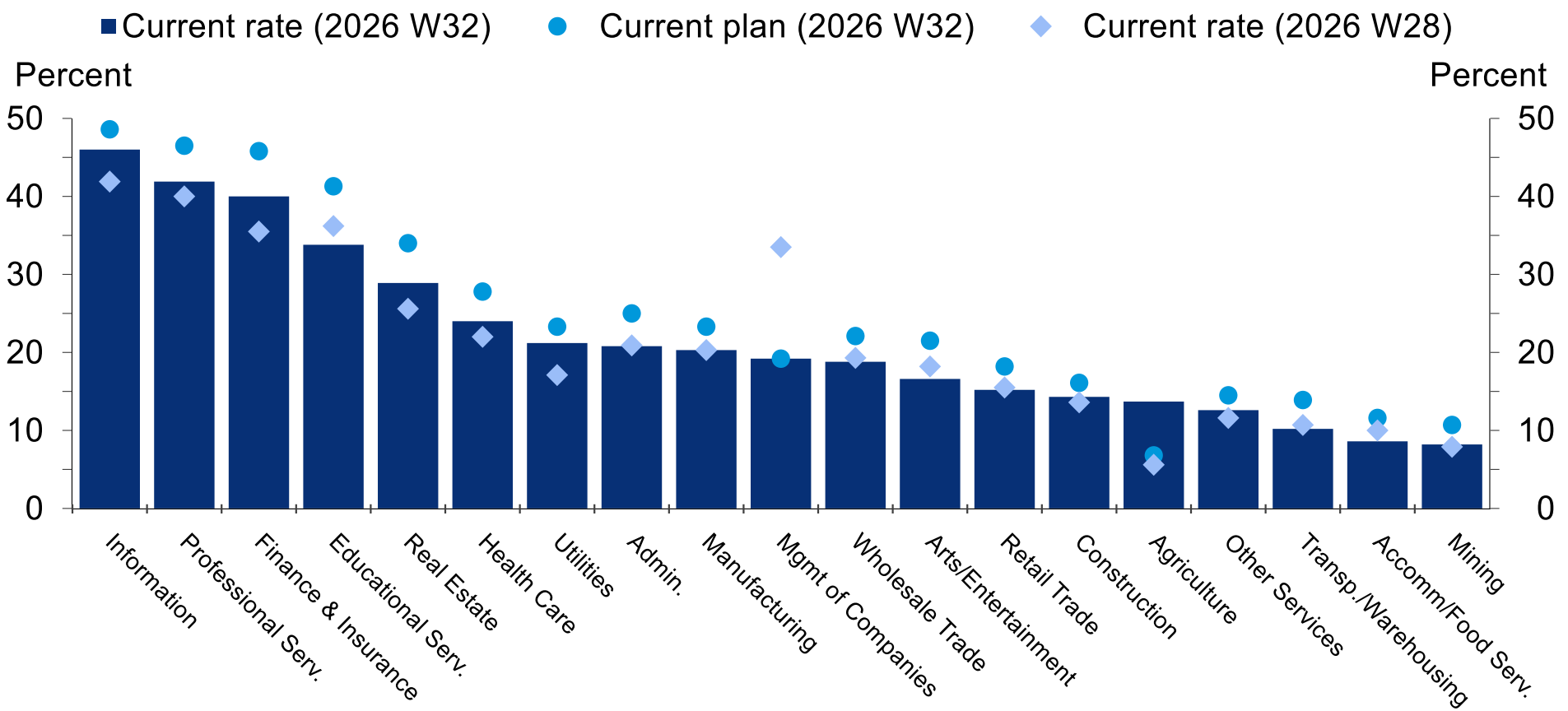


Source: Census BTOS, Deutsche Bank

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Significant variation in AI adoption rates across sectors according to Census data

AI adoption rate

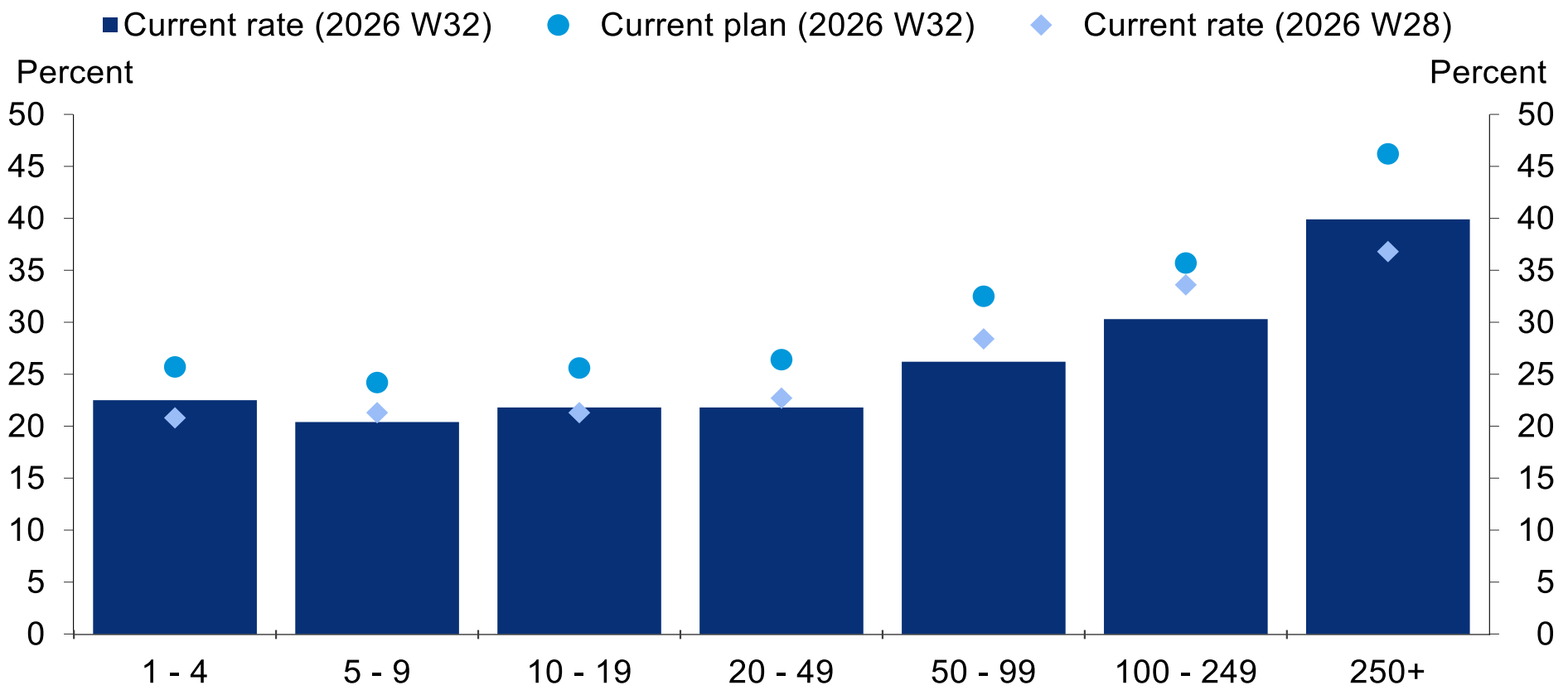


Source: USCB, Macrobond, Deutsche Bank

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Larger firms have higher rates of AI adoption; generally true across sectors as well

AI adoption by firm size

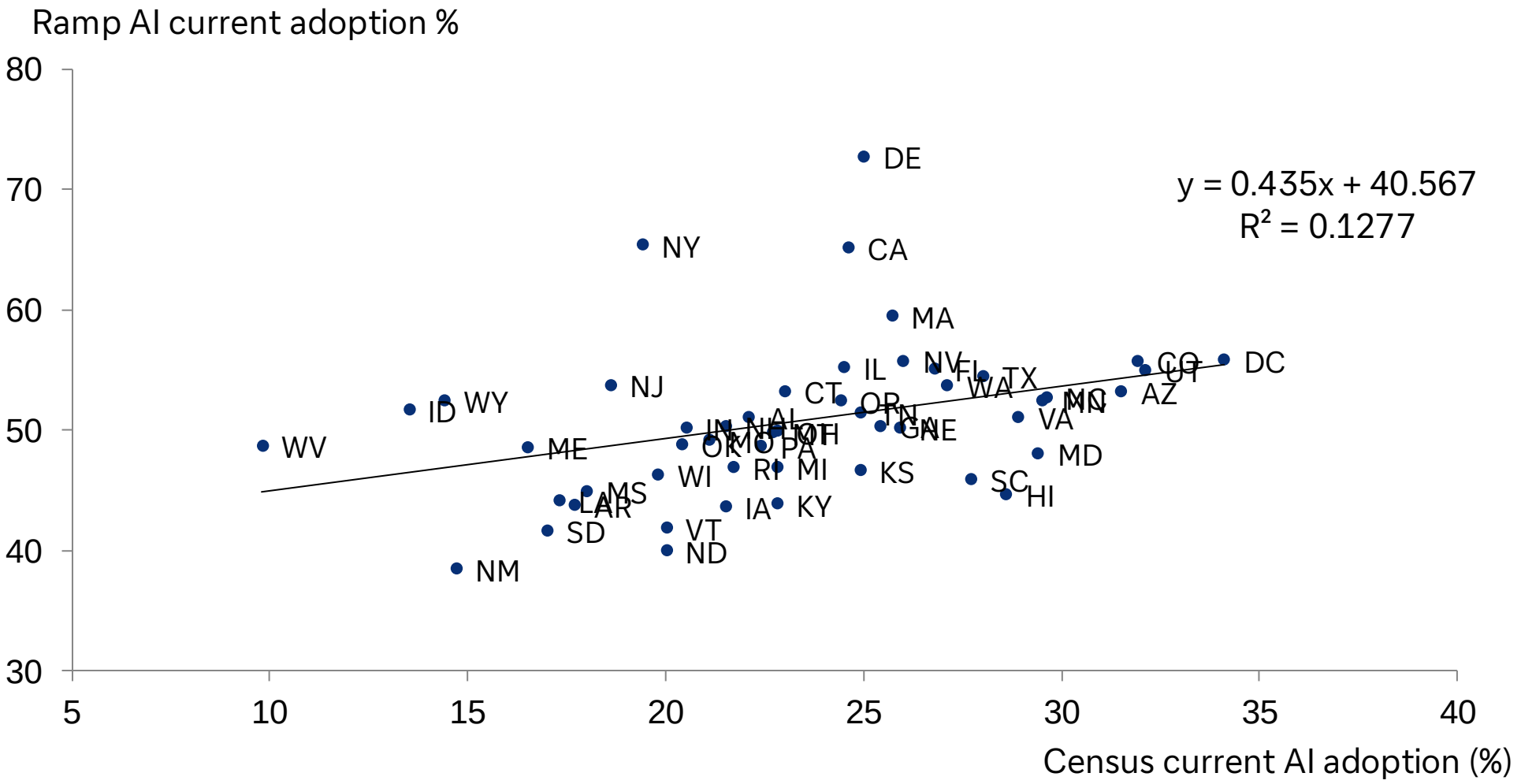


Source: USCB, Macrobond, Deutsche Bank

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Measures of AI adoption rates across states broadly agree but there is a good amount of variation

AI adoption rates across states

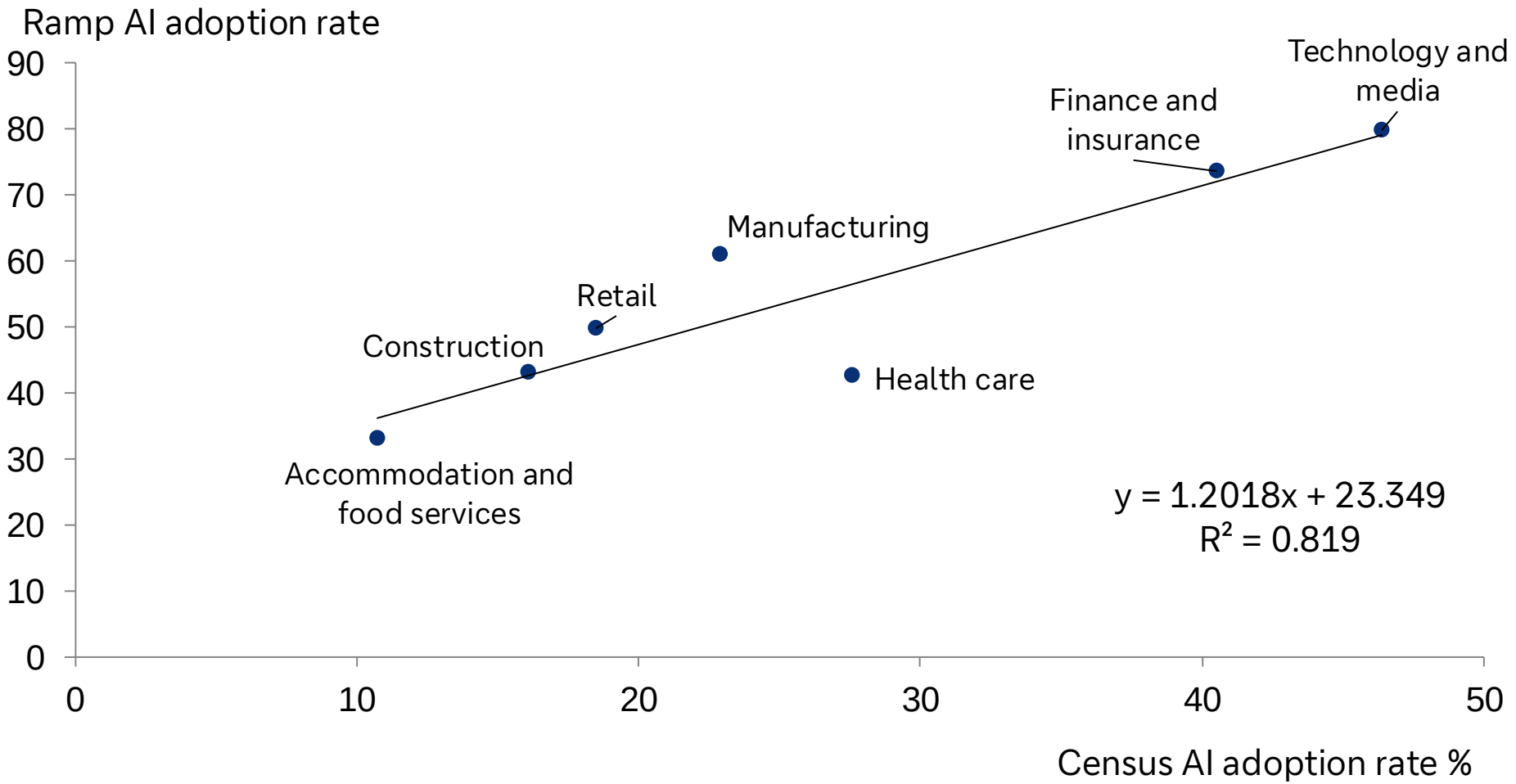


Source: Ramp, Census BTOS, Deutsche Bank

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Agreement across AI adoption rate measures at the sector level

AI adoption rate across sectors

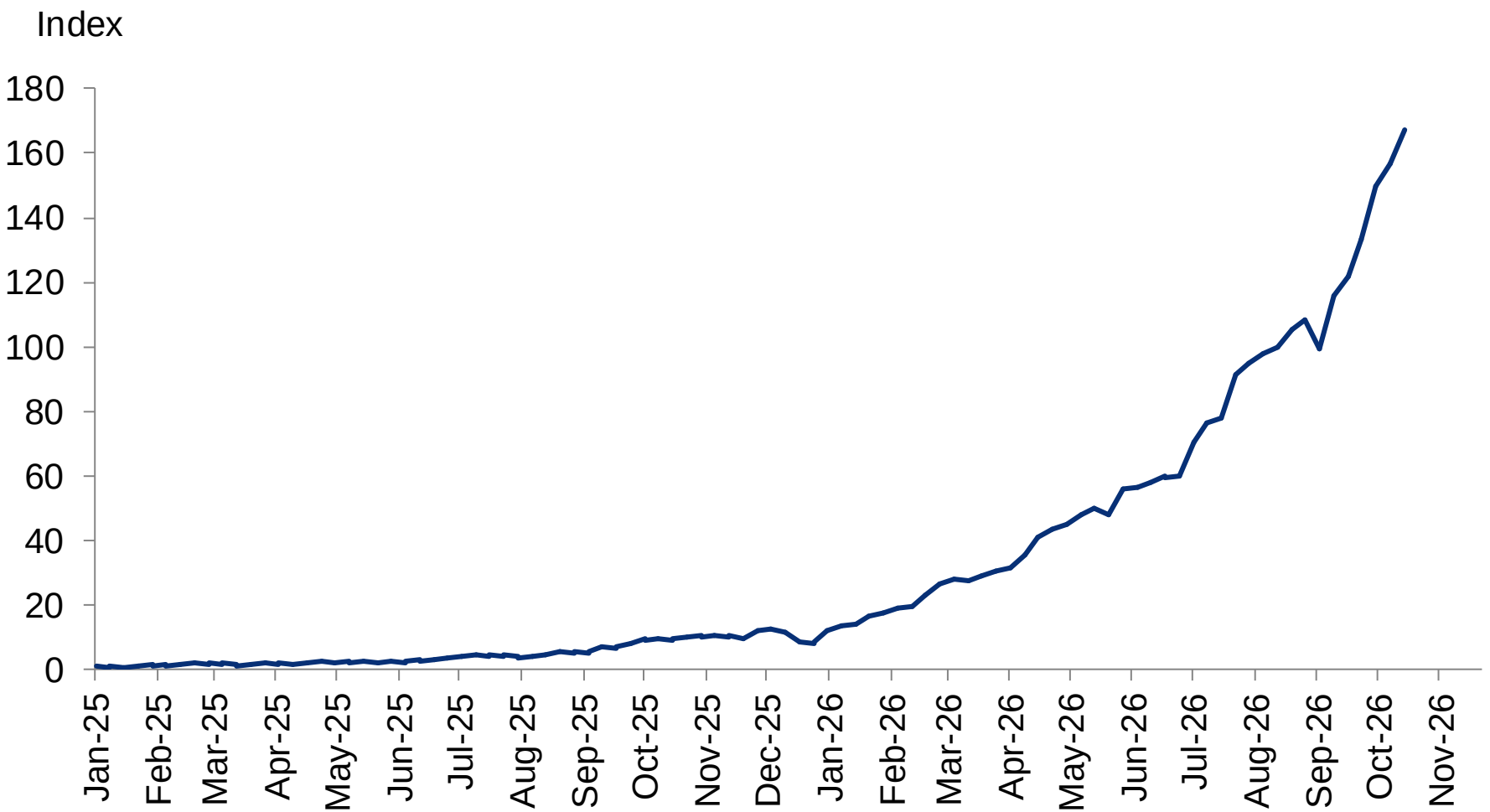


Source: Ramp, Census BTOS, Deutsche Bank

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Total token usage has continued to rise sharply

Top AI models total token usage

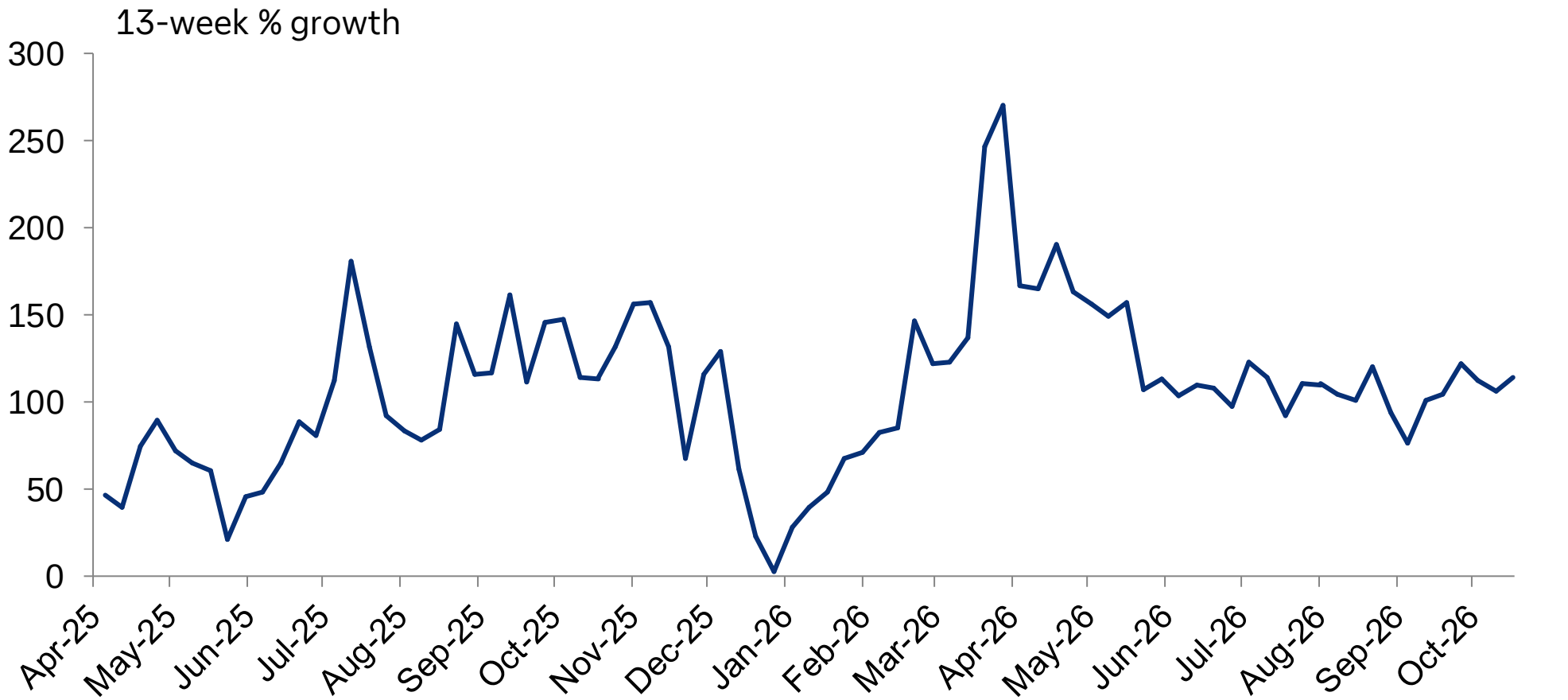


Source: Ramp, Deutsche Bank

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Growth rate of total token usage has been stable near 100% over a quarterly horizon

Top AI models total usage

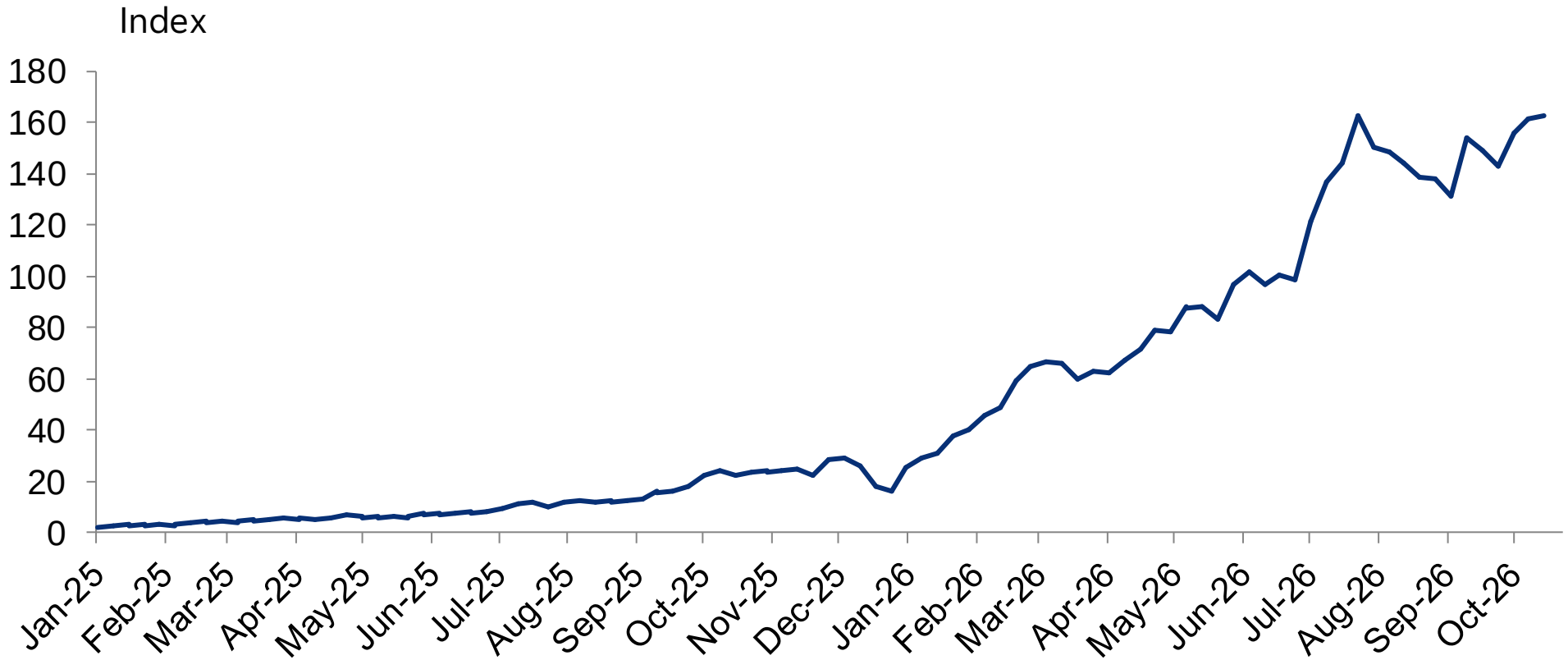


Source: Ramp, Deutsche Bank

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Total token spend has stabilized in recent months after rising sharply through H1 2026

Top AI models total spend volume index

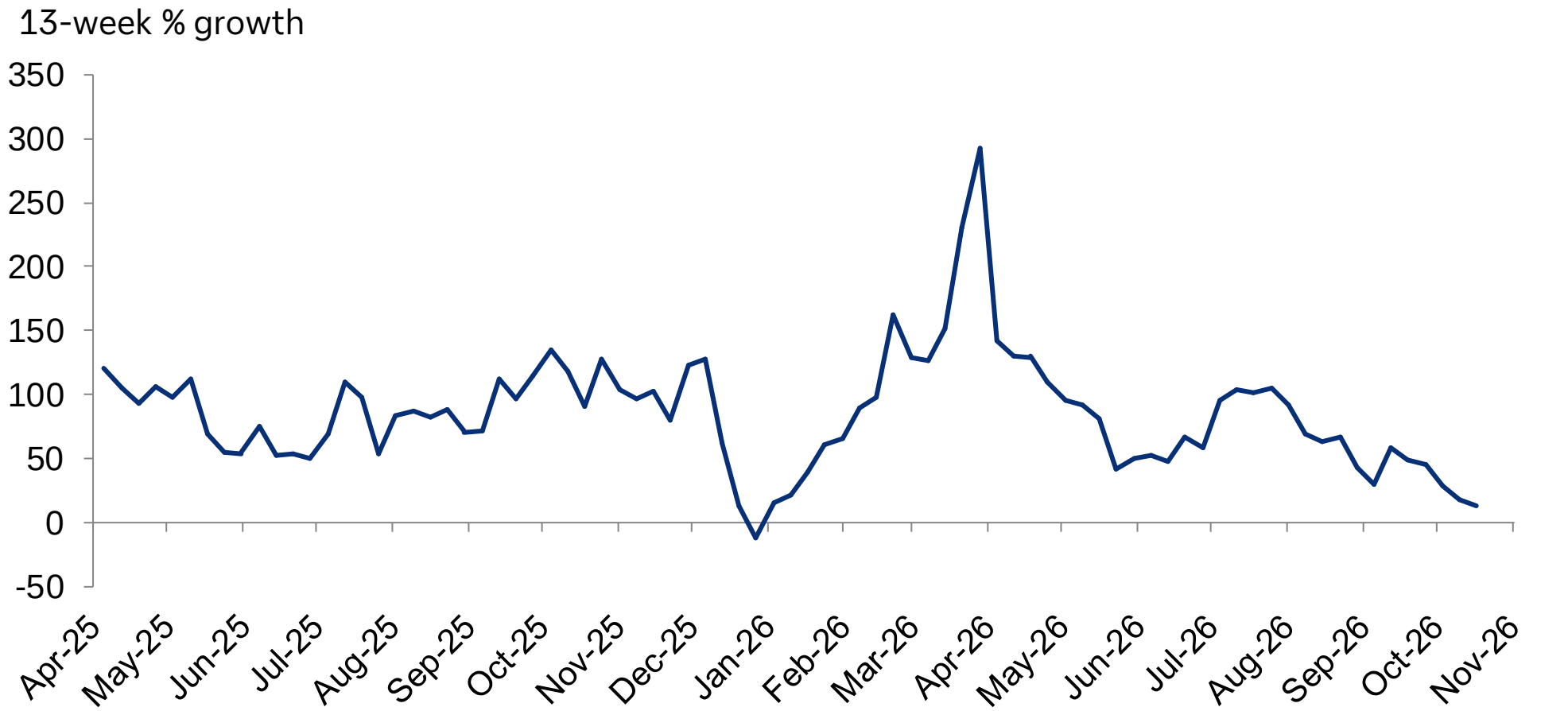


Source: Ramp, Deutsche Bank

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Quarterly growth rate of AI spending has been decelerating

Top AI models total spend volume index

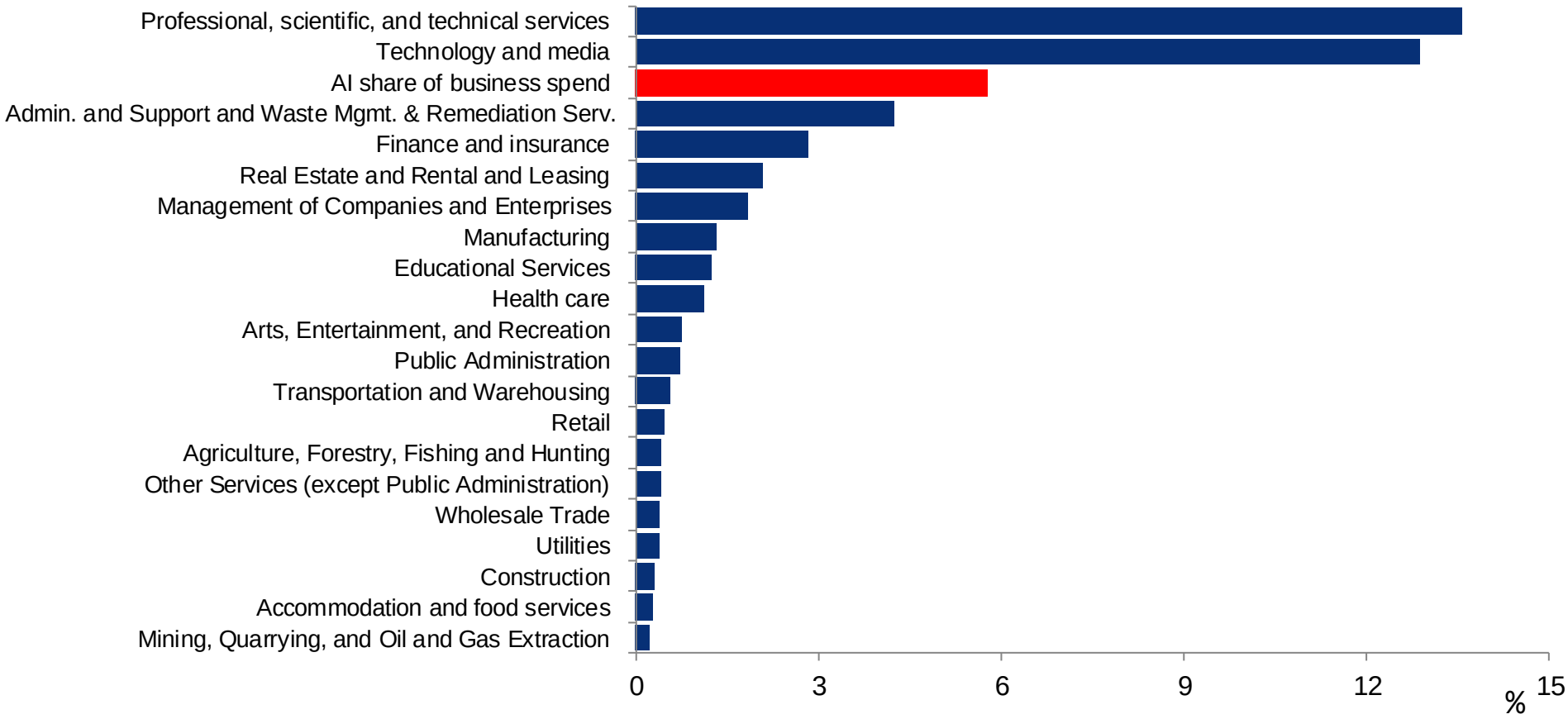


Source: Ramp, Deutsche Bank

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AI spending as share of overall spending ex-payroll

AI share of business spend excluding payroll



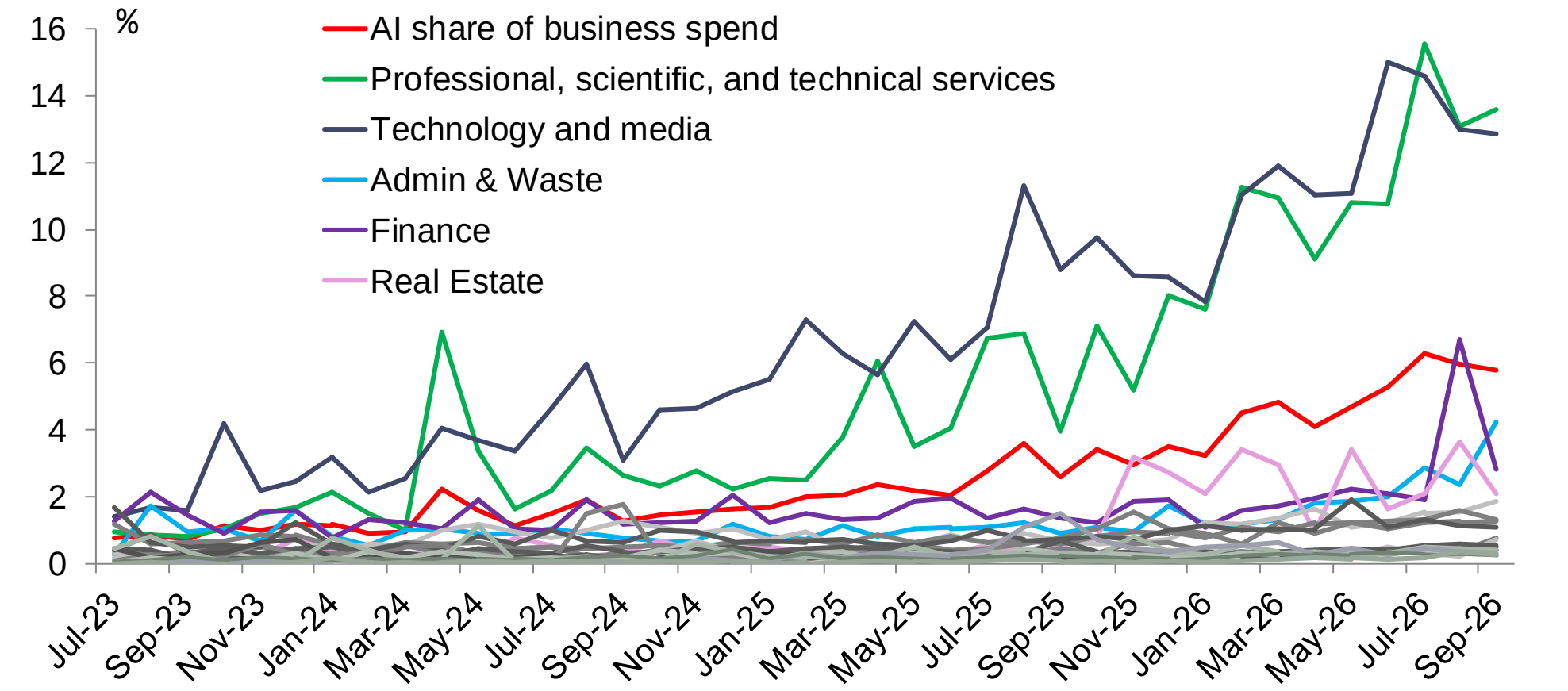
Business spend data from Ramp includes corporate card and invoice / ACH payments. AI spend includes LLM subscriptions, coding agent subscriptions, API tokens, and GPU cloud and infrastructure spend.

Source: Ramp, Deutsche Bank

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AI spend has risen sharply as a share of total business spend but is still low on an absolute basis

AI business spend as share of total spend



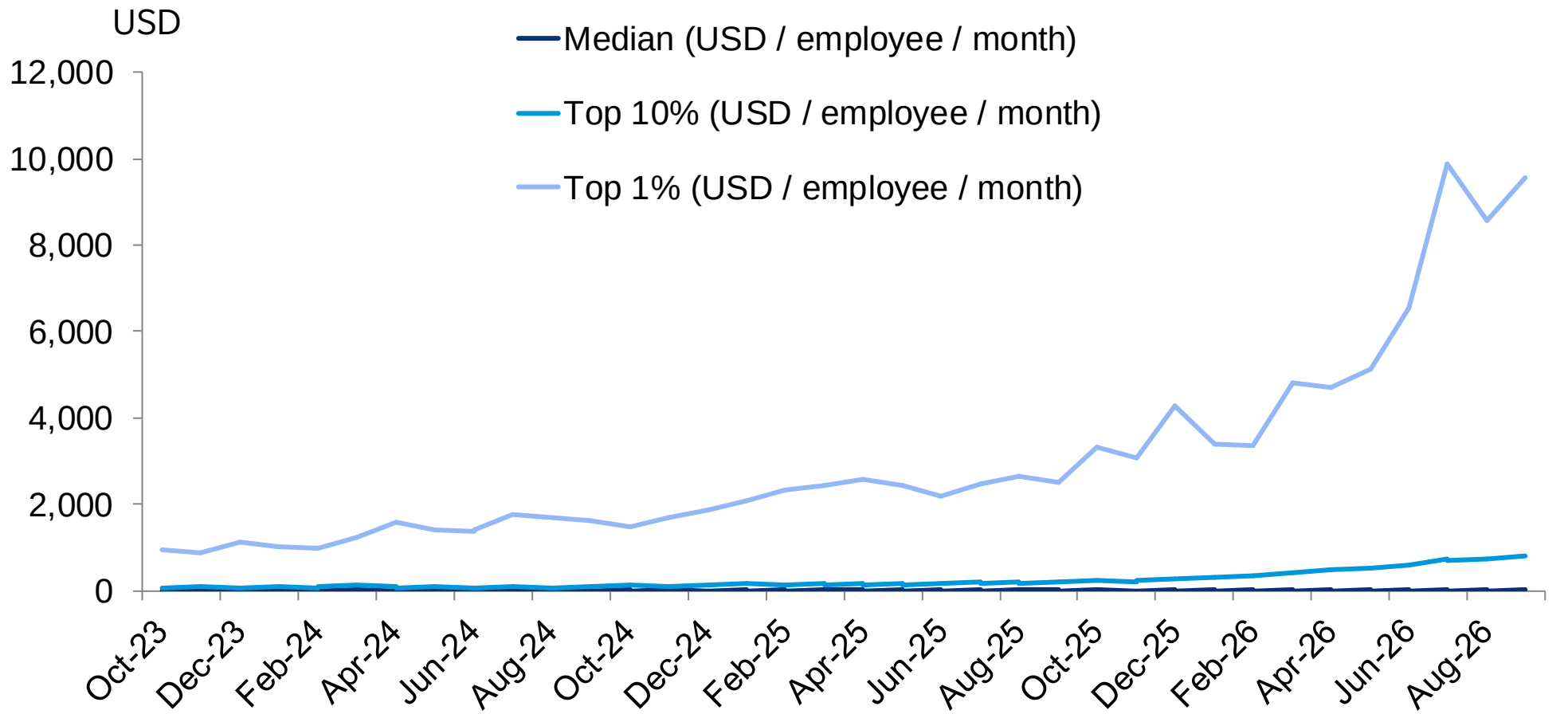
Note: Business spend data from Ramp includes corporate card and invoice / ACH payments. AI spend includes LLM subscriptions, coding agent subscriptions, API tokens, and GPU cloud and infrastructure spend.

Source: Ramp, Deutsche Bank

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AI spending is heavily skewed towards high users – top 1% spends nearly $10k per employee per month, median is $16

Monthly AI spend per employee



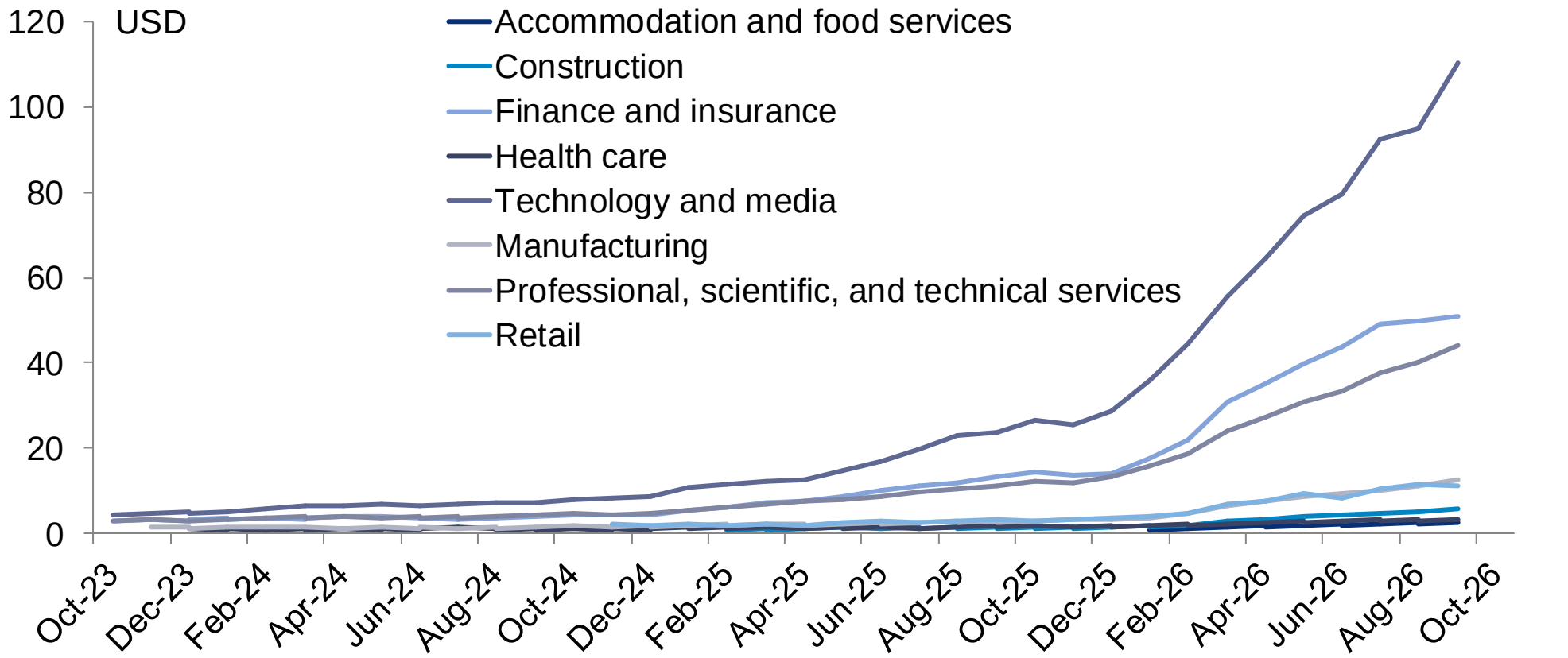
Note: Business spend data from Ramp includes corporate card and invoice / ACH payments. AI spend includes LLM subscriptions, coding agent subscriptions, API tokens, and GPU cloud and infrastructure spend

Source: Ramp, Deutsche Bank

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AI spend is also heavily skewed towards the tech industry – median AI spend per employee 2x as large in tech vs. finance

Median monthly AI spend per employee by sector



Business spend data from Ramp includes corporate card and invoice / ACH payments. AI spend includes LLM subscriptions, coding agent subscriptions, API tokens, and GPU cloud and infrastructure spend

Source: Ramp, Deutsche Bank

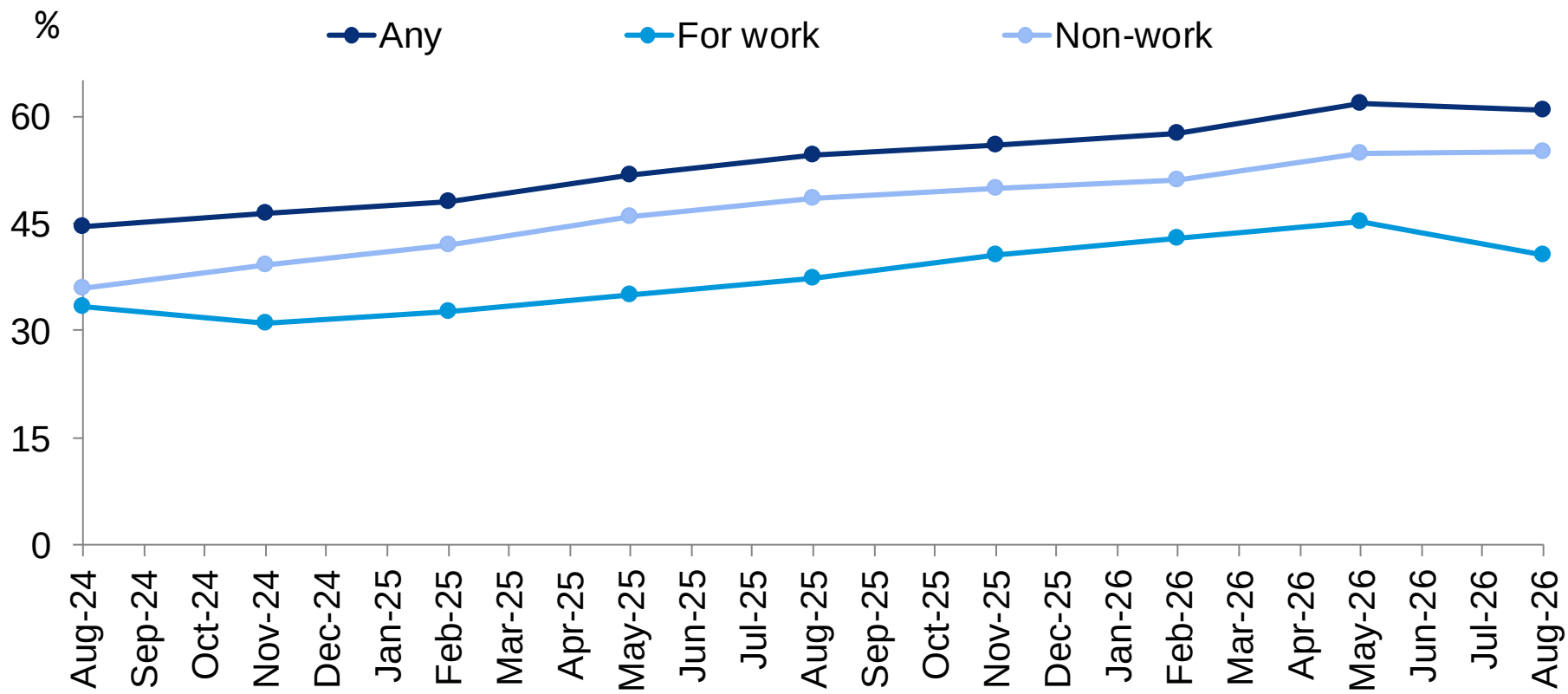
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3.
Worker and consumer adoption

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Share of workers using AI has risen significantly over the past two years (45%->60%); work usage recently slipped

Share of working age adults using GenAI

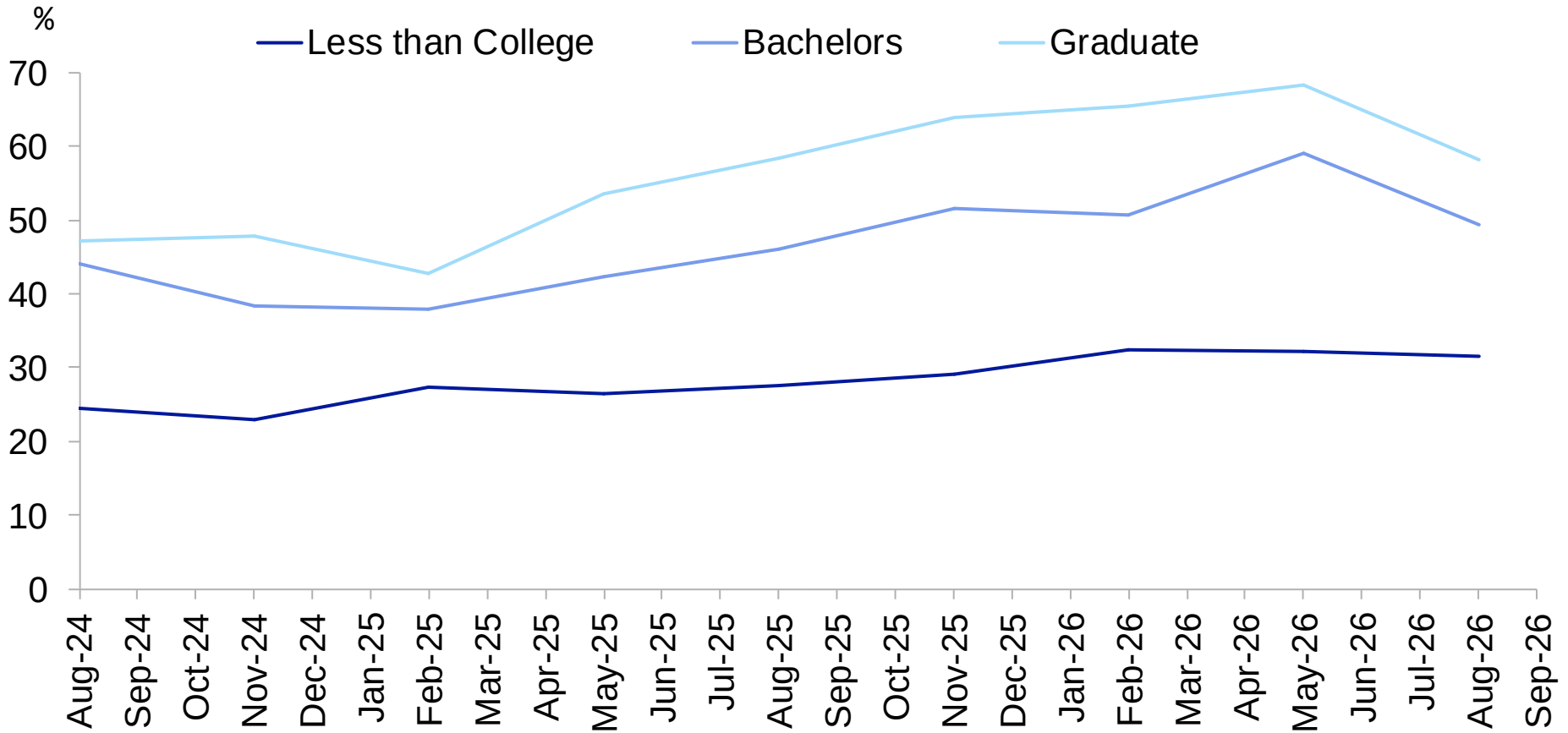


Source: Harvard Kennedy School Project on Workforce & Federal Reserve Bank of St. Louis, GenAI Adoption Tracker (Real-Time Population Survey), Deutsche Bank

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AI usage is positively related to educational attainment

Percentage of employed respondents who use GenAI for their job at any frequency

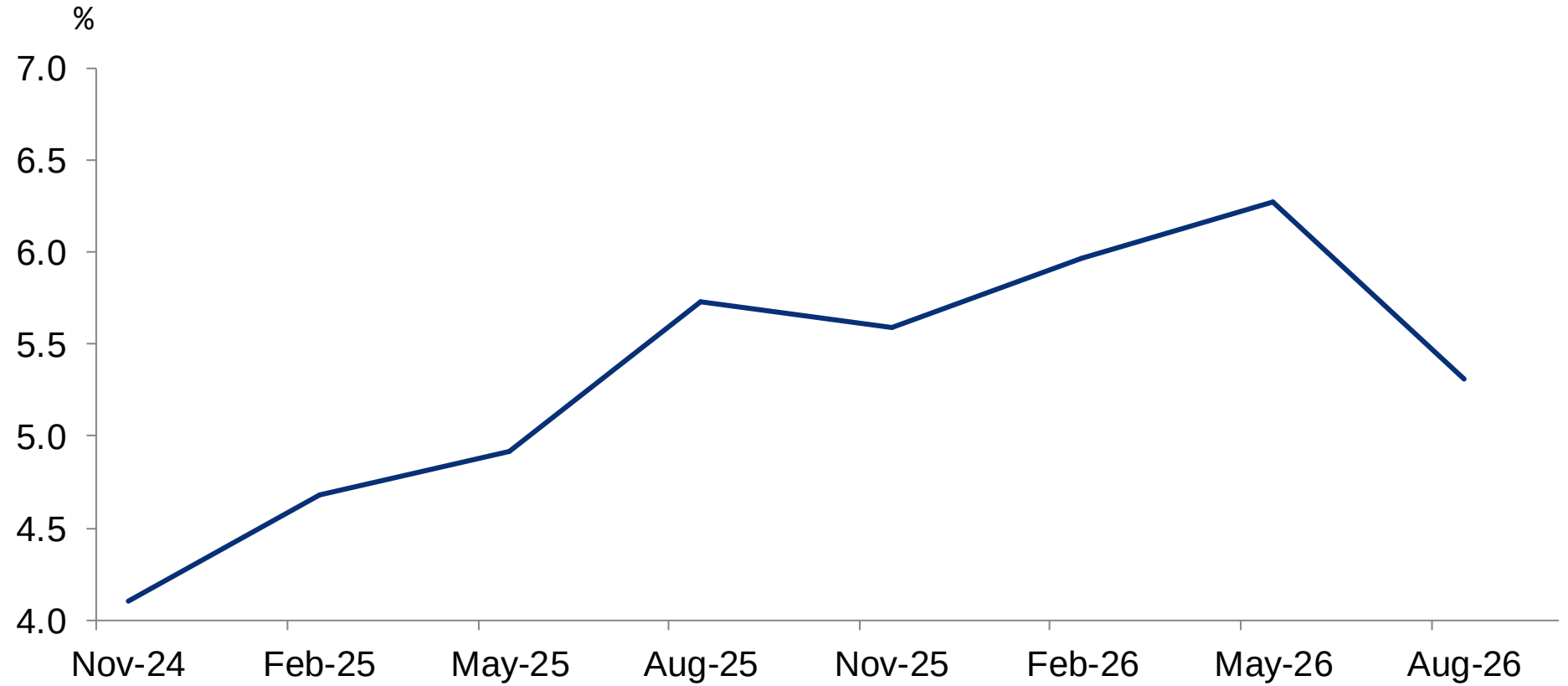


Source: Harvard Kennedy School Project on Workforce & Federal Reserve Bank of St. Louis, GenAI Adoption Tracker (Real-Time Population Survey), Deutsche Bank

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Share of workweek using GenAI trending upward but recently declined

Overall percentage of employed workers' weekly work hours spent actively using GenAI for work

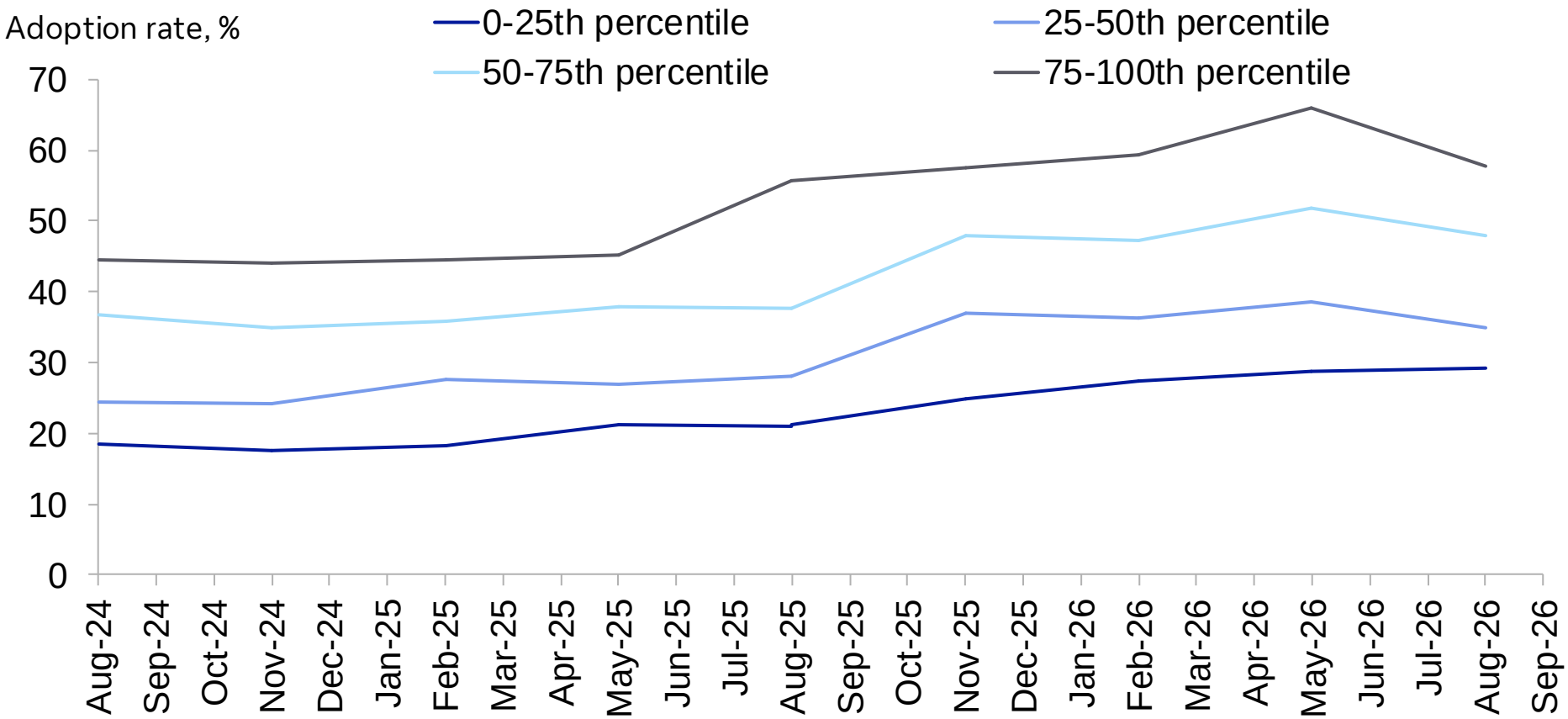


Source: Harvard Kennedy School Project on Workforce & Federal Reserve Bank of St. Louis, GenAI Adoption Tracker (Real-Time Population Survey), Deutsche Bank

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AI usage positively related to earnings

Share of adults ages 18–64 who use GenAI for their work: by earnings quartile

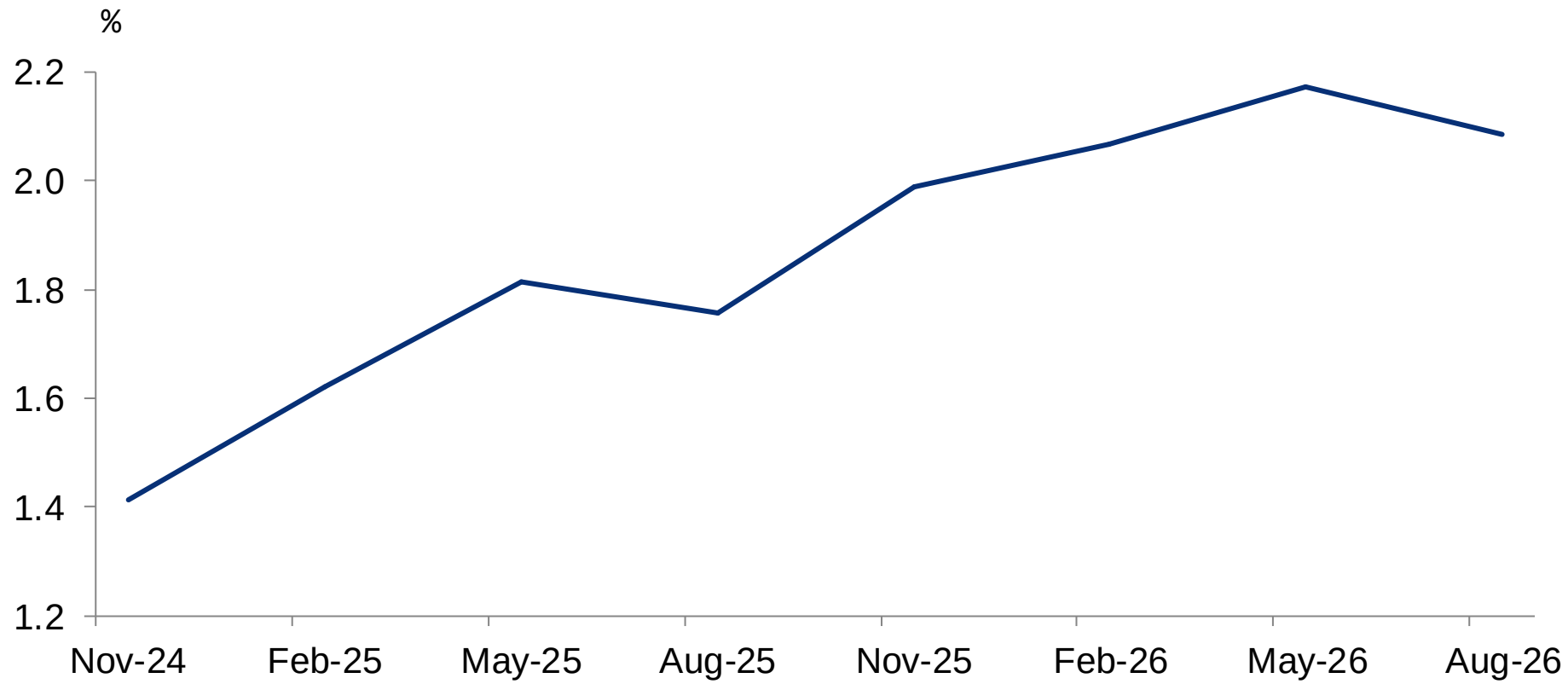


Source: Harvard Kennedy School Project on Workforce & Federal Reserve Bank of St. Louis, GenAI Adoption Tracker (Real-Time Population Survey), Deutsche Bank

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Percentage of work hours saved due to GenAI on the rise but still very low, only ~2%

Percentage of work hours saved through workplace GenAI

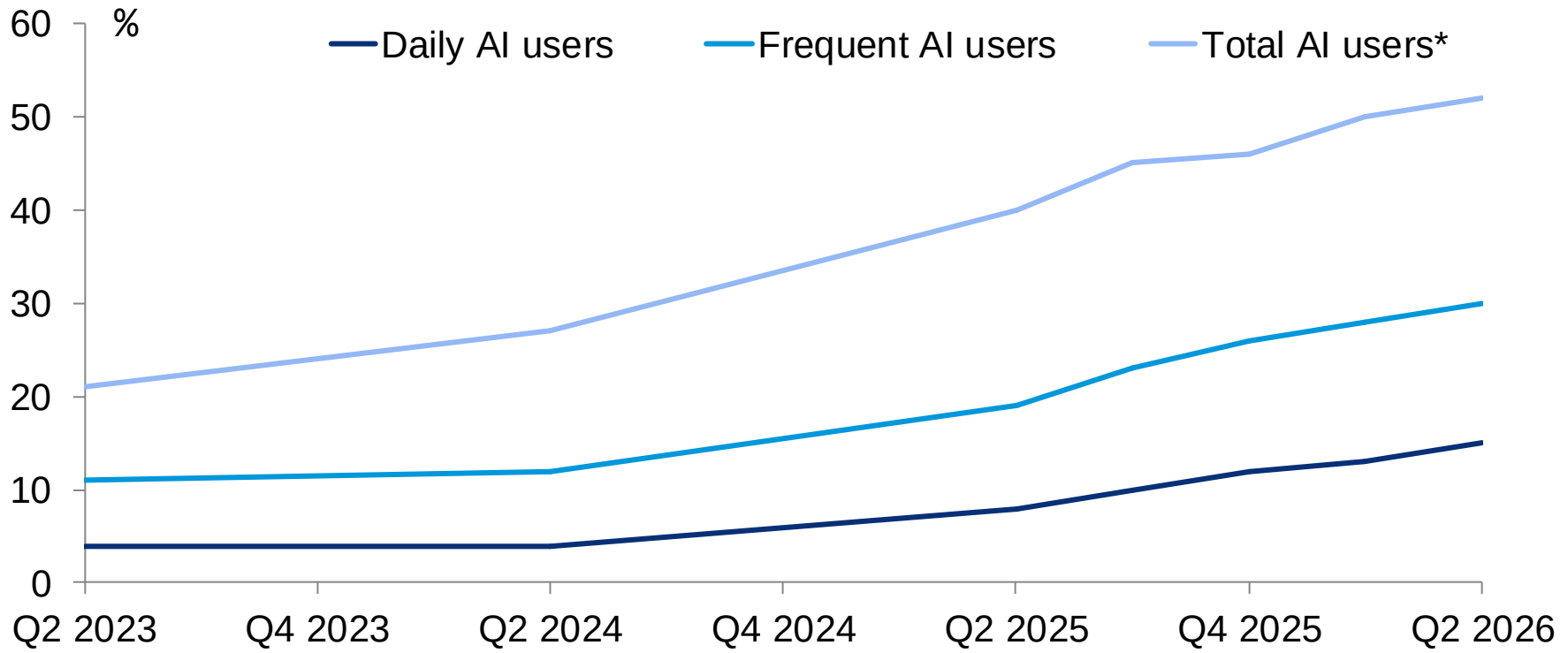


Source: Harvard Kennedy School Project on Workforce & Federal Reserve Bank of St. Louis, GenAI Adoption Tracker (Real-Time Population Survey), Deutsche Bank

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AI usage by frequency

Individual AI use at work



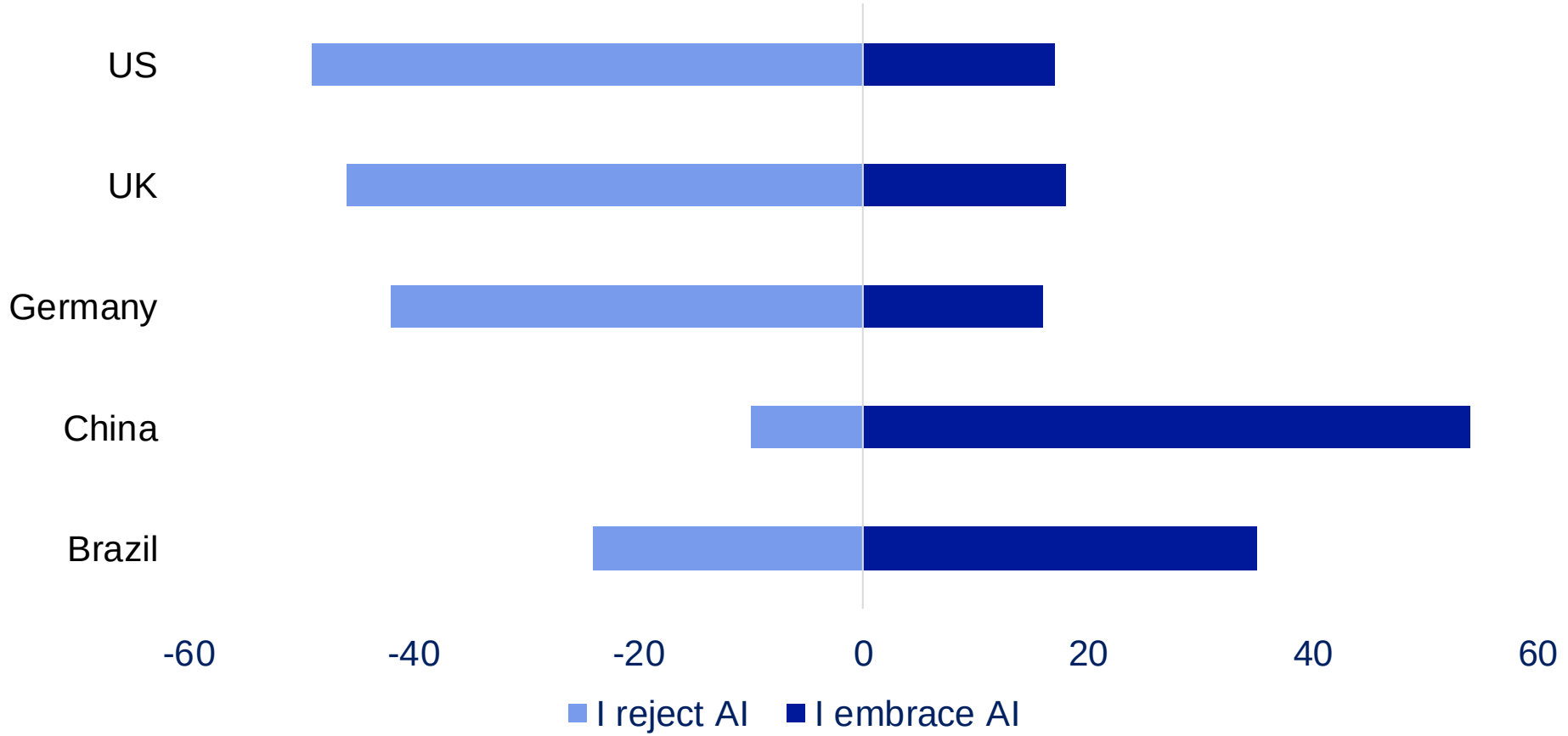
**Note: % using AI daily, a few times a week, a few times a month or a few times a year*

Source: Harvard Kennedy School Project on Workforce & Federal Reserve Bank of St. Louis, GenAI Adoption Tracker (Real-Time Population Survey), Deutsche Bank

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Trust in AI in the US and other developed markets is fragile compared with markets such as China and Brazil

Survey participants in select countries (%)

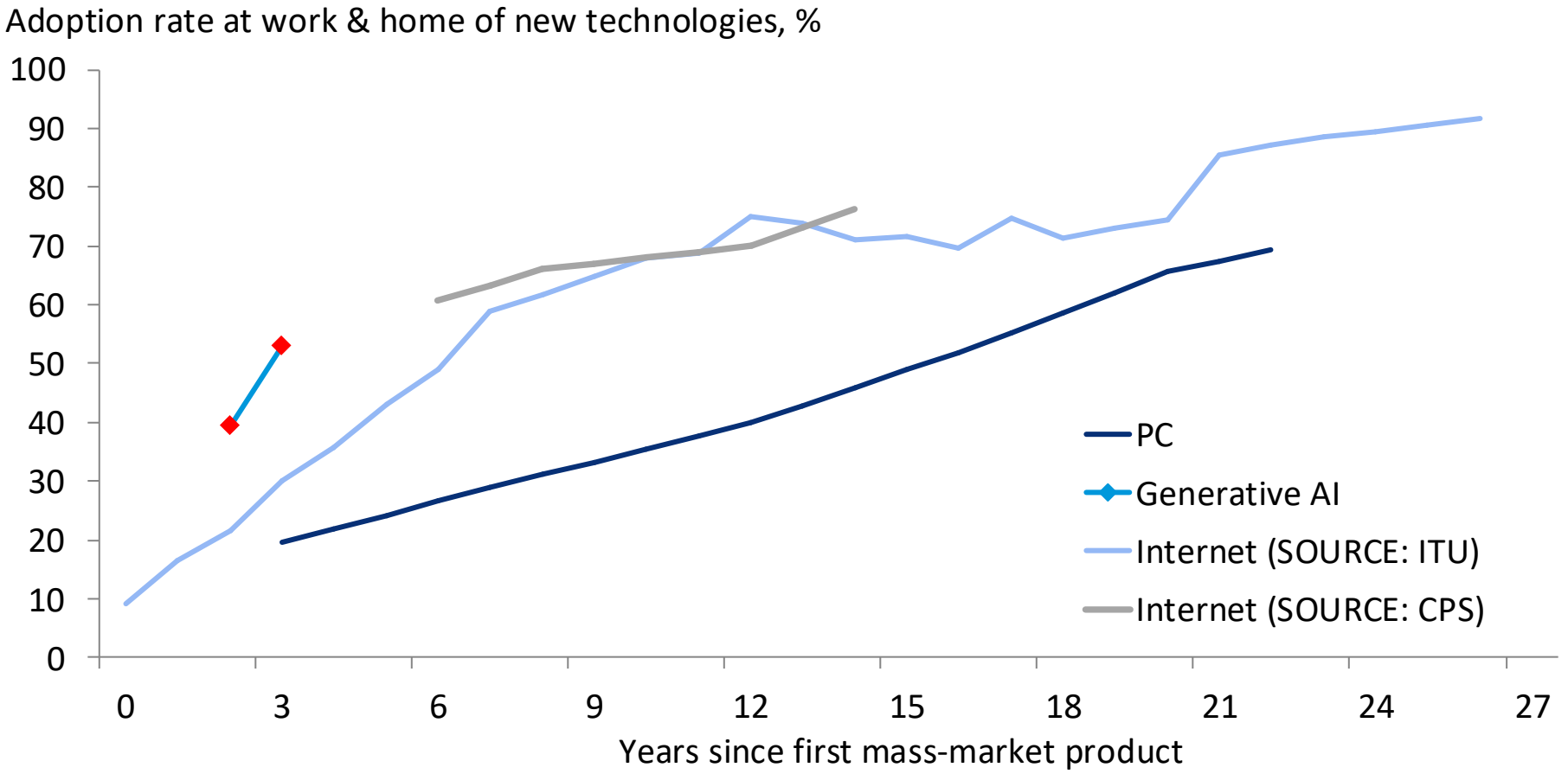


Source: Edelman Trust Barometer Flash Poll: Trust and AI at a Crossroads (2025), Deutsche Bank

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AI adoption outpacing historical technological revolutions

Adoption rates of historical technologies



Source: Bick, Blandin & Deming, Real-Time Population Survey, CPS, ITU, Stanford AI Report 2026, Deutsche Bank

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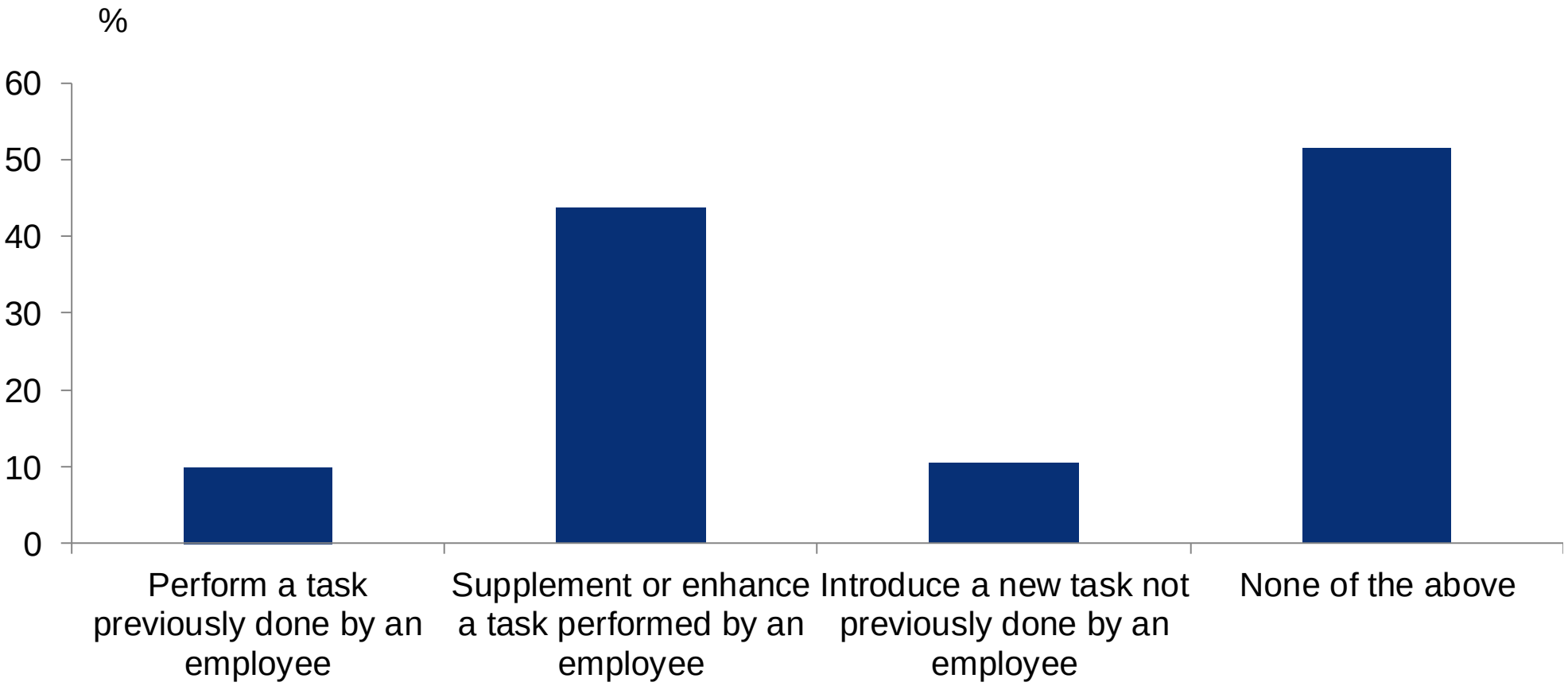
4.
How AI is being used



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Firms using AI are mostly using it to "supplement or enhance" employees

In the last 6 months, did this business use AI to do any of the following?



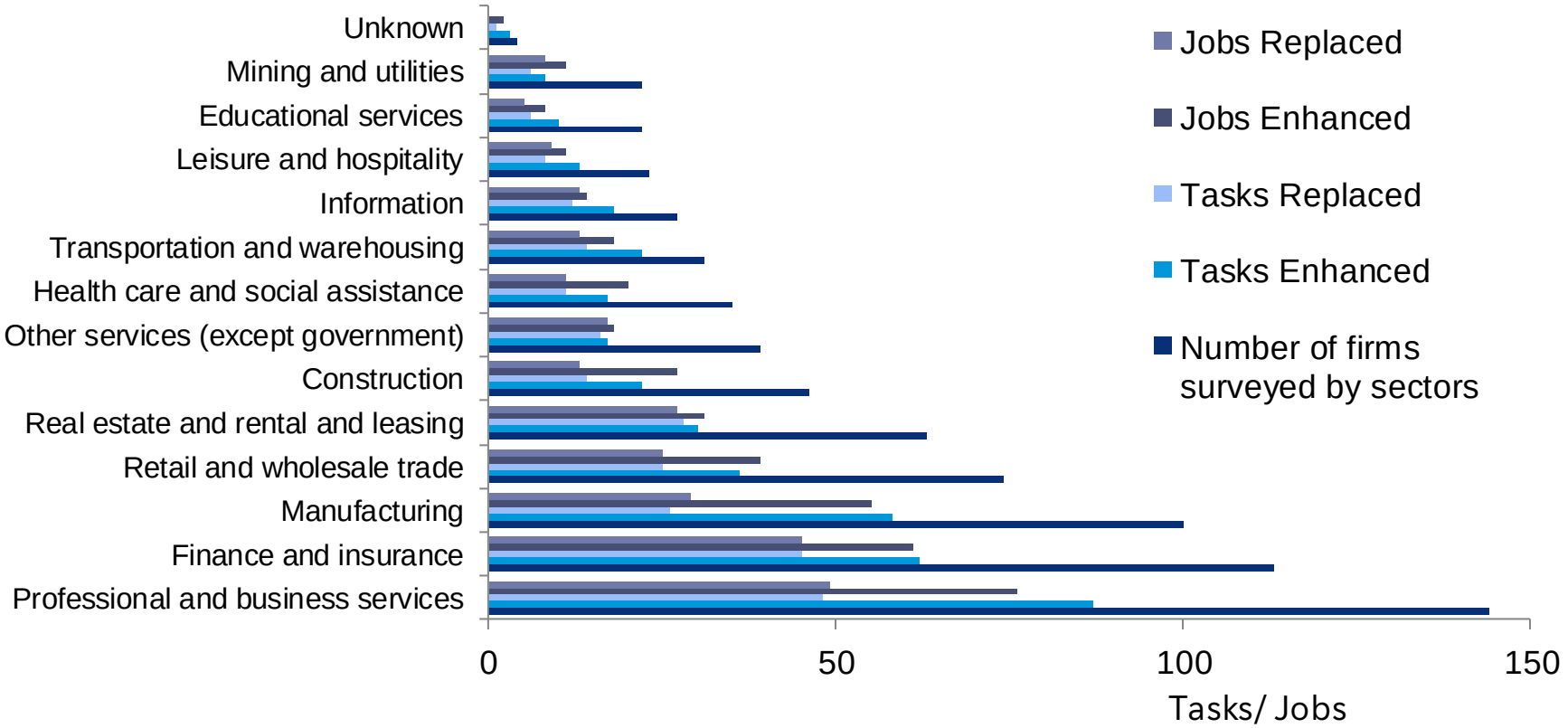
Note: Combined estimates for data collected 11/17/2025 to 02/08/2026

Source: BTOS, Census, Deutsche Bank

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Significant variation across sectors in how AI is used relative to jobs and tasks

AI impact across sectors*



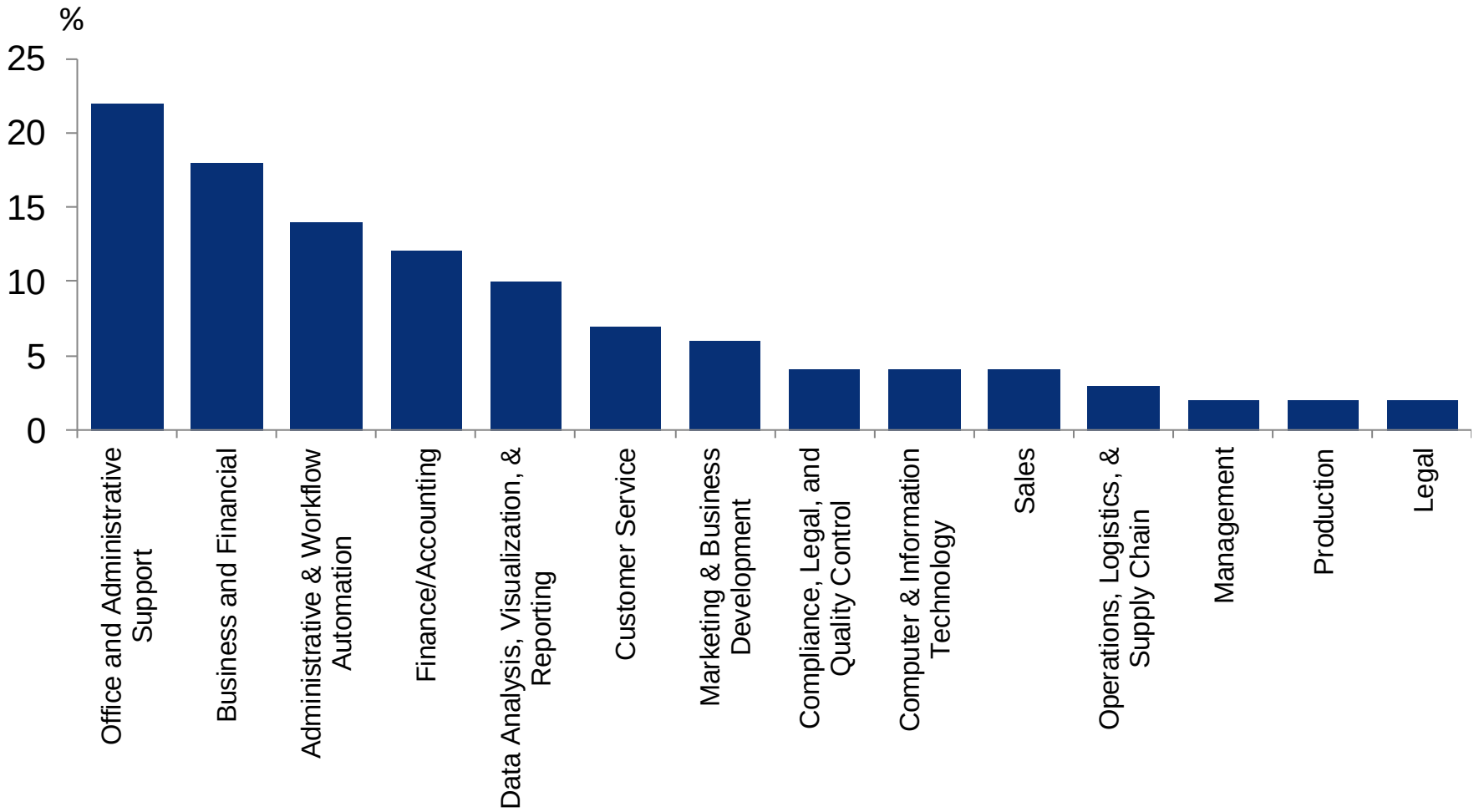
*Summary of AI impact on tasks and jobs by industry sector. Shows how many firms in each sector reported enhanced or replaced tasks/jobs.

Source: Duke University, Deutsche Bank

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Significant variation across occupations in using AI to replace tasks/jobs

Share (%) of tasks / jobs replaced by AI in sectors

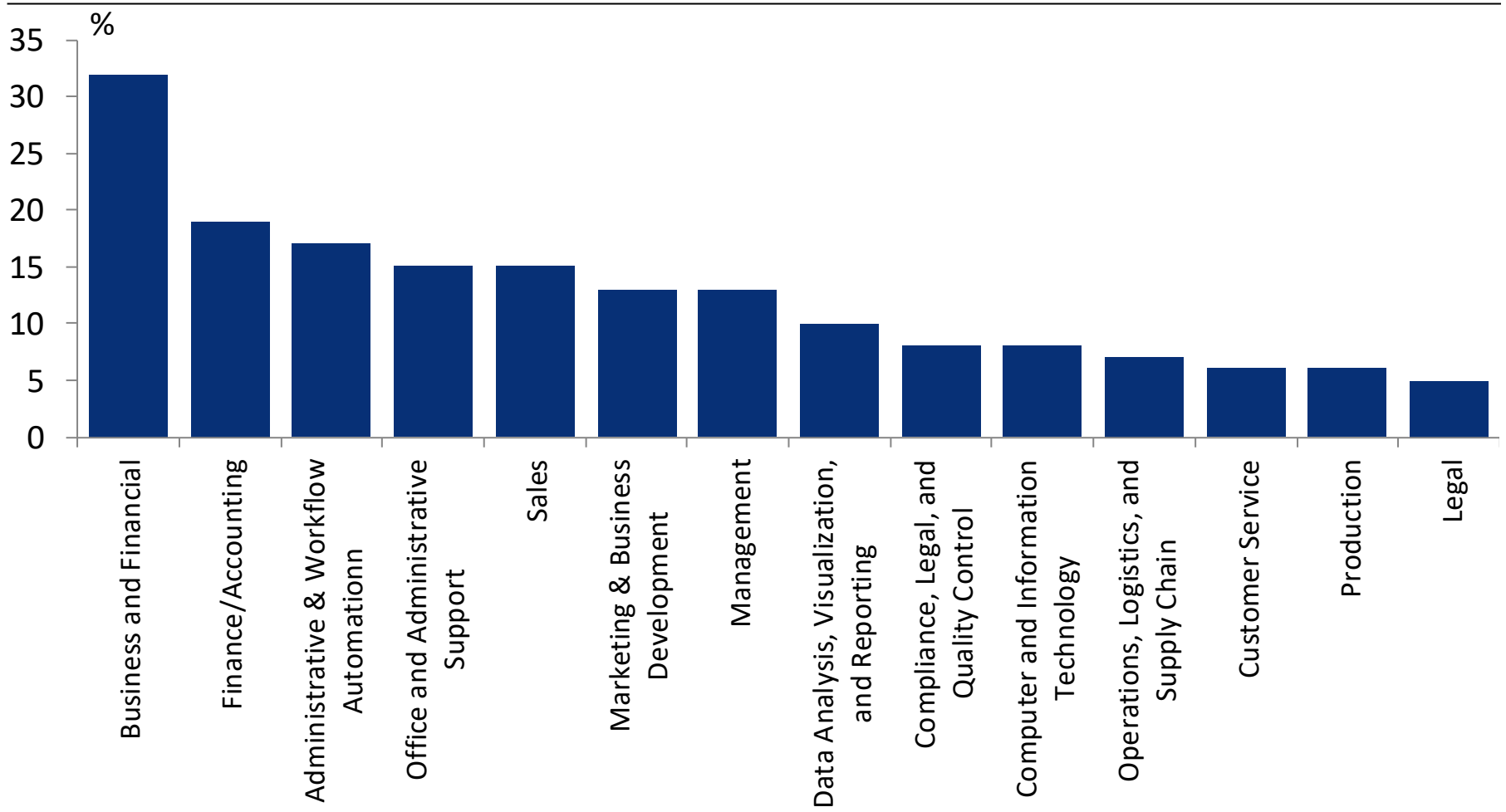


Source: Duke University, Deutsche Bank

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Significant variation across occupations in using AI to enhance tasks/jobs

Share (%) of tasks / jobs enhanced by AI in sectors

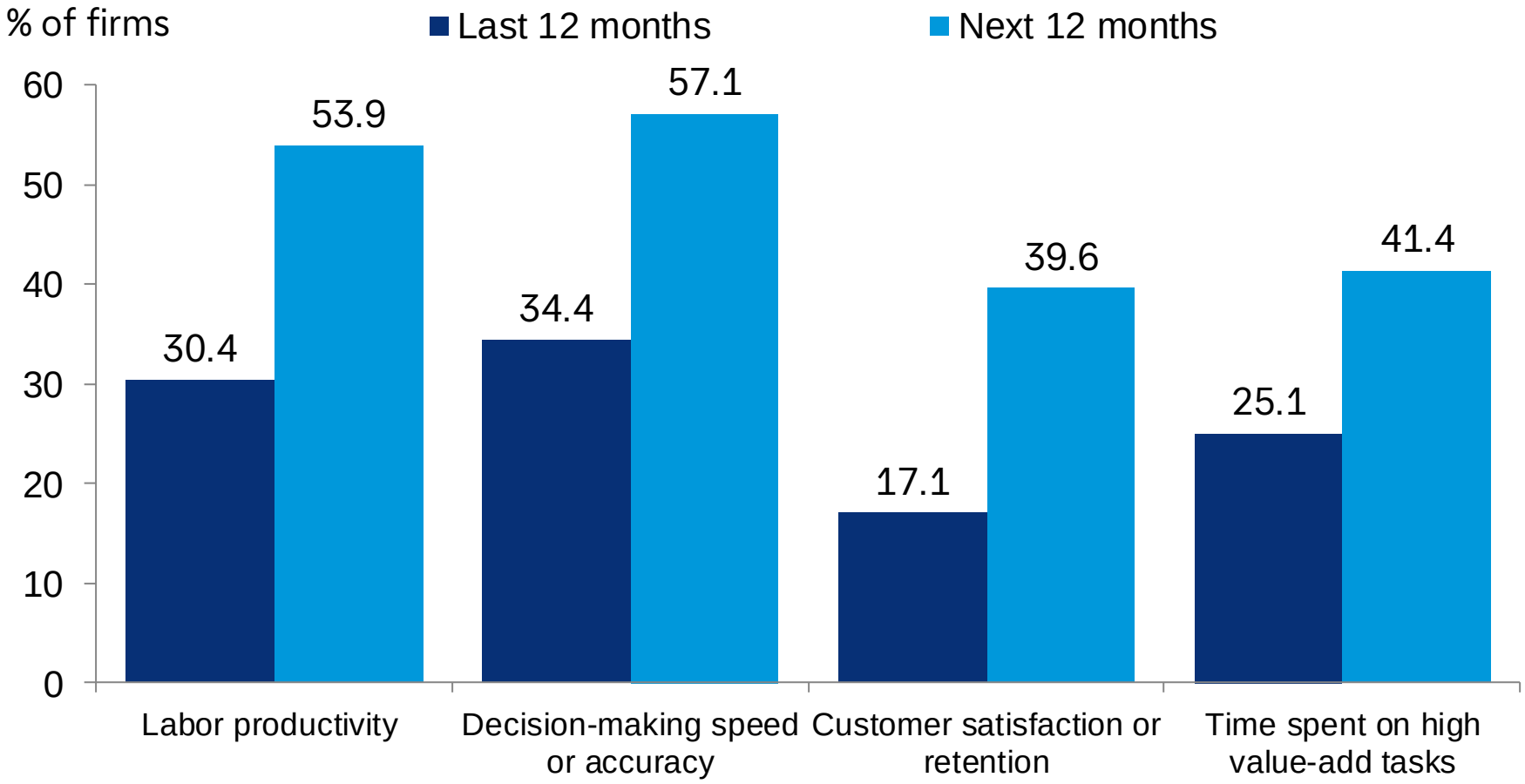


Source: Duke University, Deutsche Bank

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Firms are optimistic that AI will begin to improve labor productivity over the next twelve months

Efficency and outcomes for firms due to use of AI: Net balance



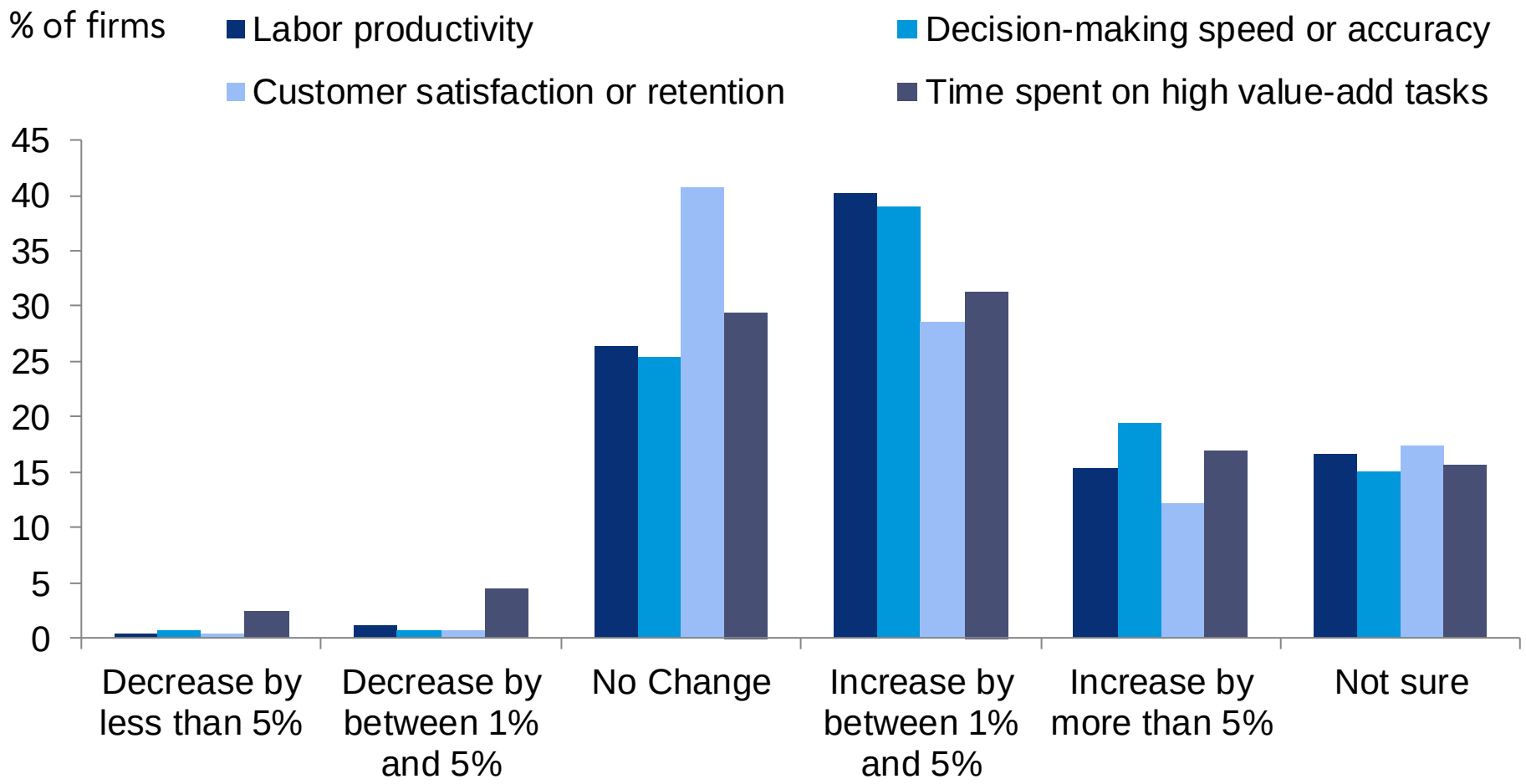
Note: Sample sizes differ slightly between the last 12 months and the next 12 months

Source: Duke University, Richmond Fed, Atlanta Fed, The CFO survey (Q4 2025), Deutsche Bank

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AI expected to improve efficiency and outcomes over the next 12 months

Efficency and outcomes for firms over next 12 months due to use of AI

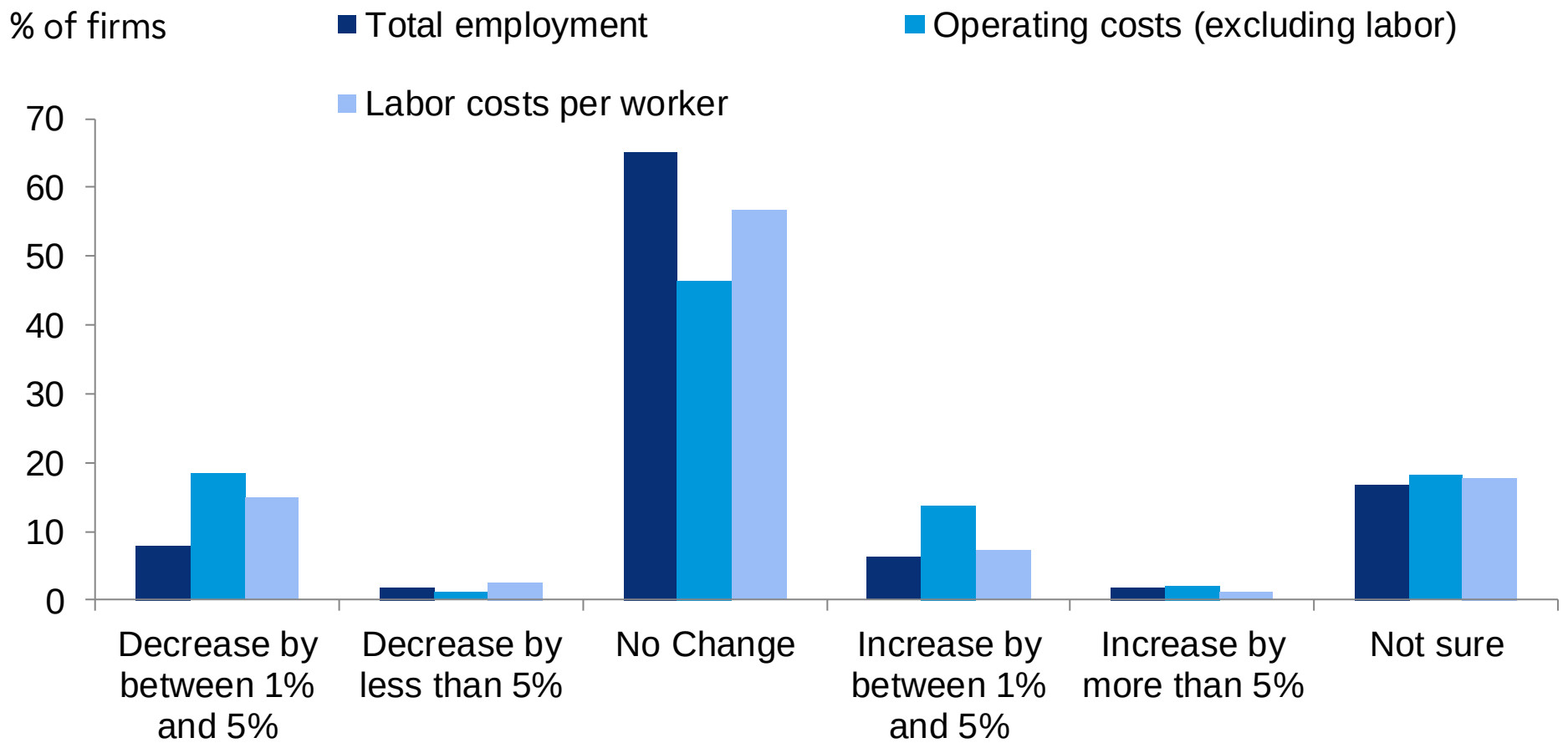


Source: Duke University, Richmond Fed, Atlanta Fed, The CFO survey (Q4 2025), Deutsche Bank

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Firms split on how AI will impact costs and employment over the next twelve months

Cost and employment for firms over next 12 months due to use of AI



Source: Duke University, Richmond Fed, Atlanta Fed, The CFO survey (Q4 2025), Deutsche Bank

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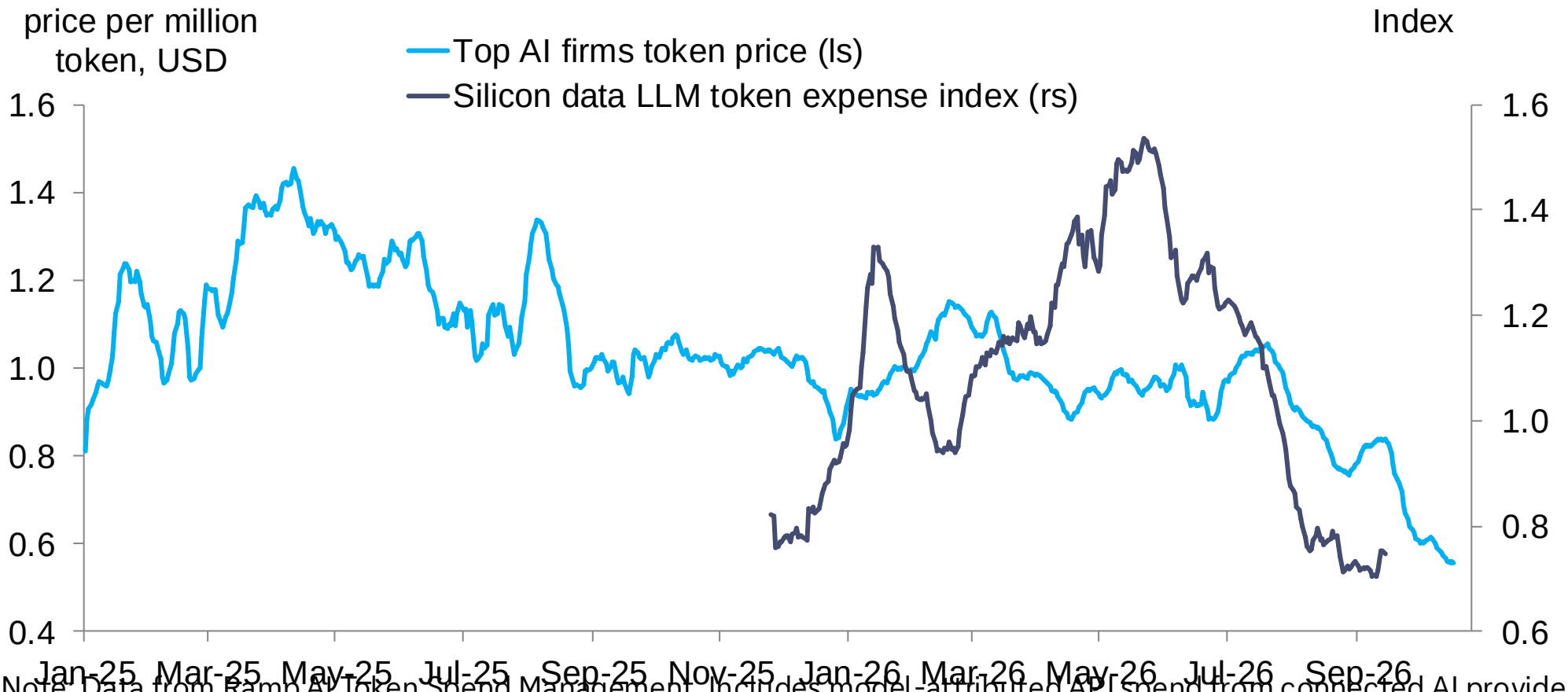
5. Pricing



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AI token prices have been trending lower in recent months

AI token prices



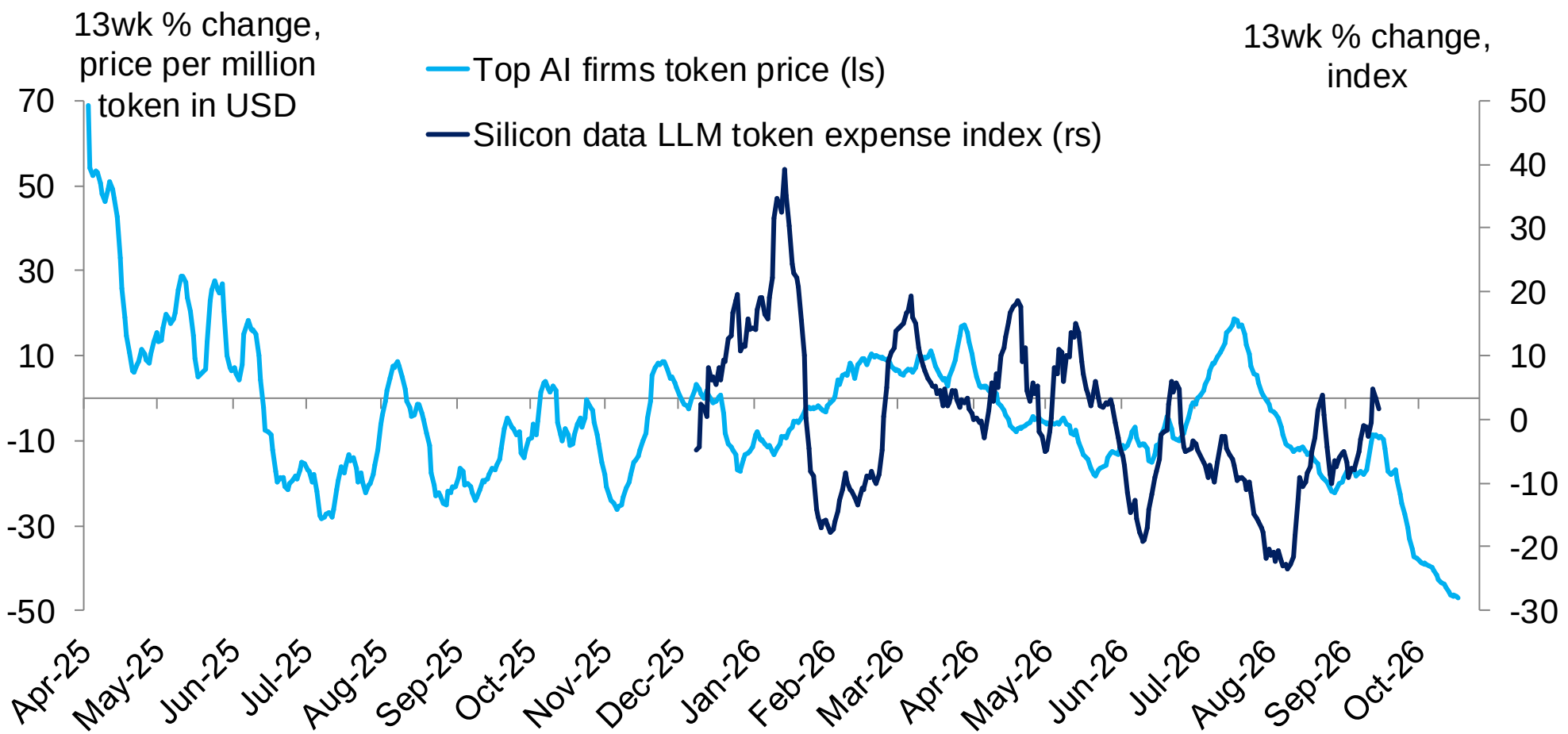
Note: Data from Ramp AI Token Spend Management. Includes model-attributed API spend from connected AI providers

Source: Ramp, Bloomberg Finance LP, Deutsche Bank

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Growth rate of AI token prices has continued to move into negative territory months

AI token price growth



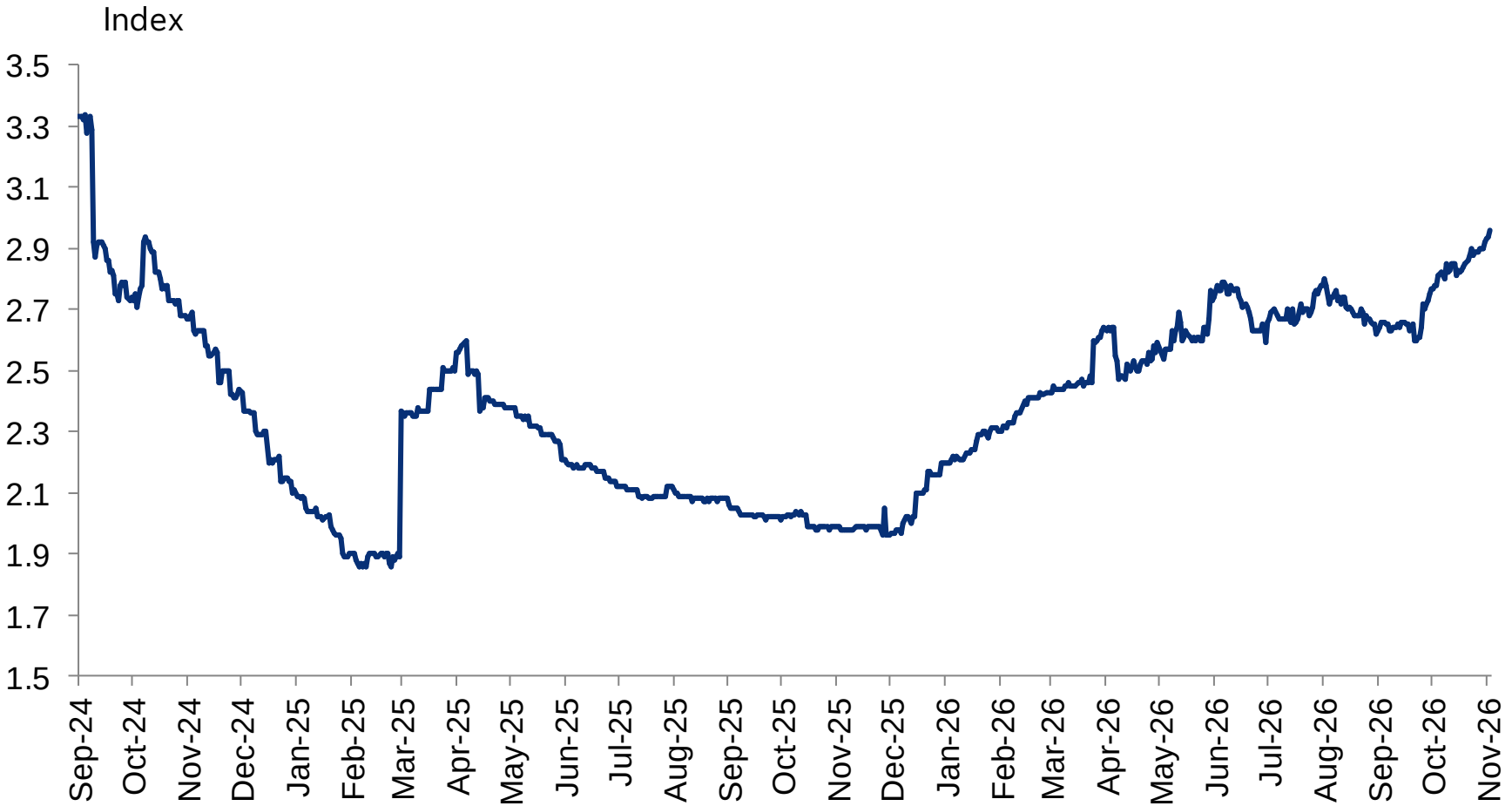
Data from Ramp AI Token Spend Management. Includes model-attributed API spend from connected AI providers

Source: Ramp, Bloomberg Finance LP, Deutsche Bank

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Chip rental prices have moved higher

Nvidia H100 chip rental index

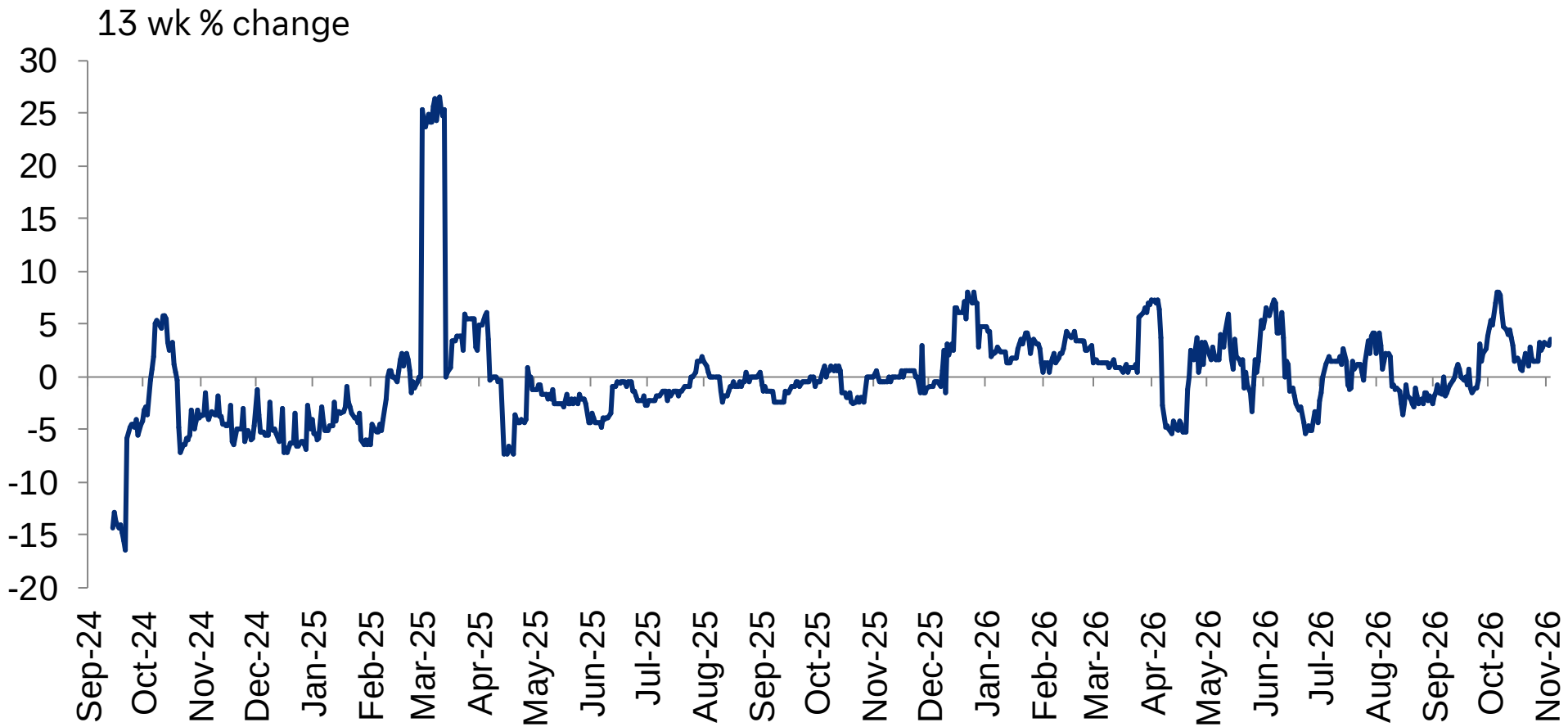


Source: Bloomberg Finance LP, Deutsche Bank

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Quarterly growth rate of chip rental prices has picked up

Nvidia H100 chip rental index



Source: Bloomberg Finance LP, Deutsche Bank

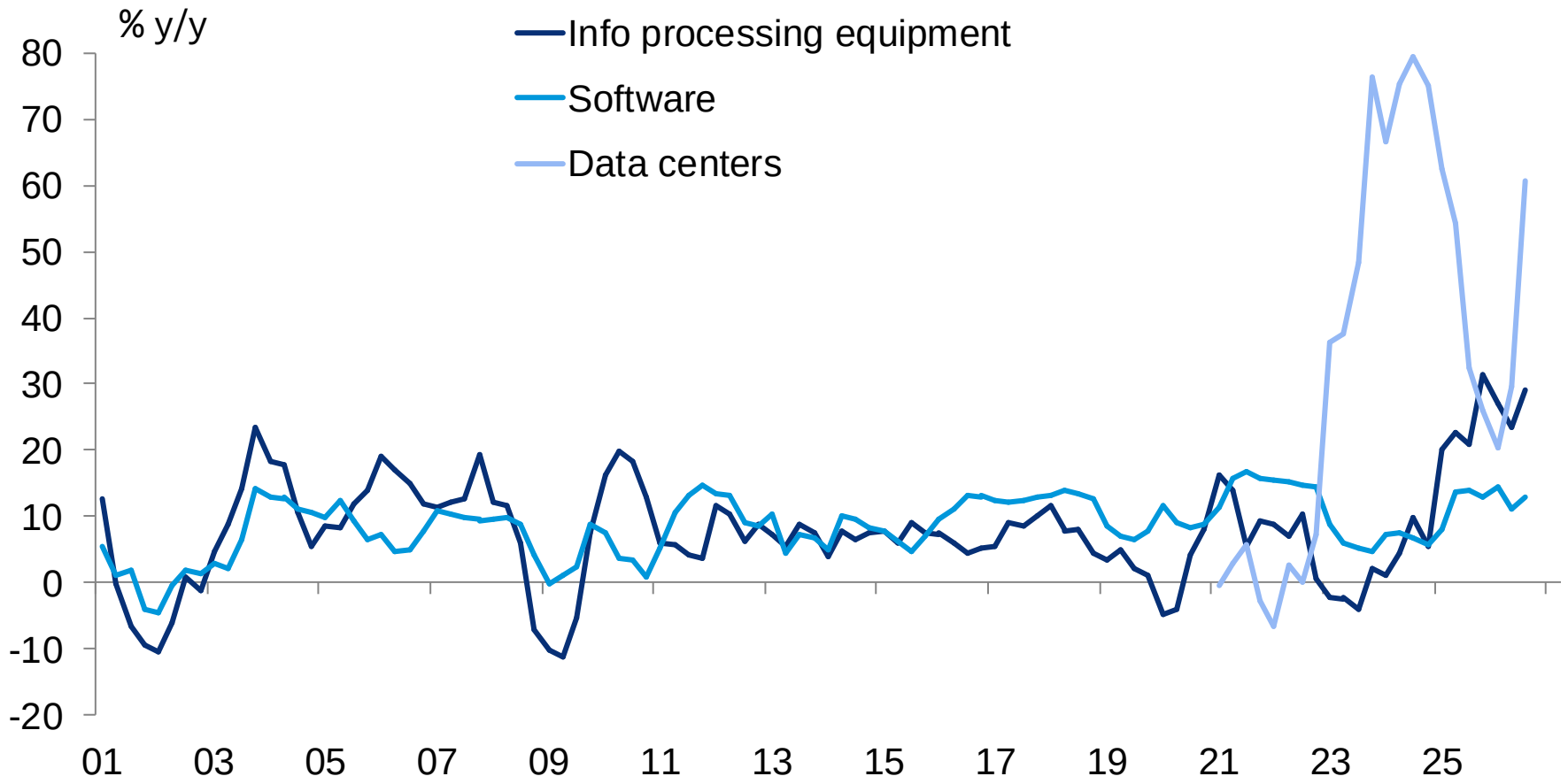
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6.
AI investment

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AI-related investment growth has surged

Growth rate of real investment spending

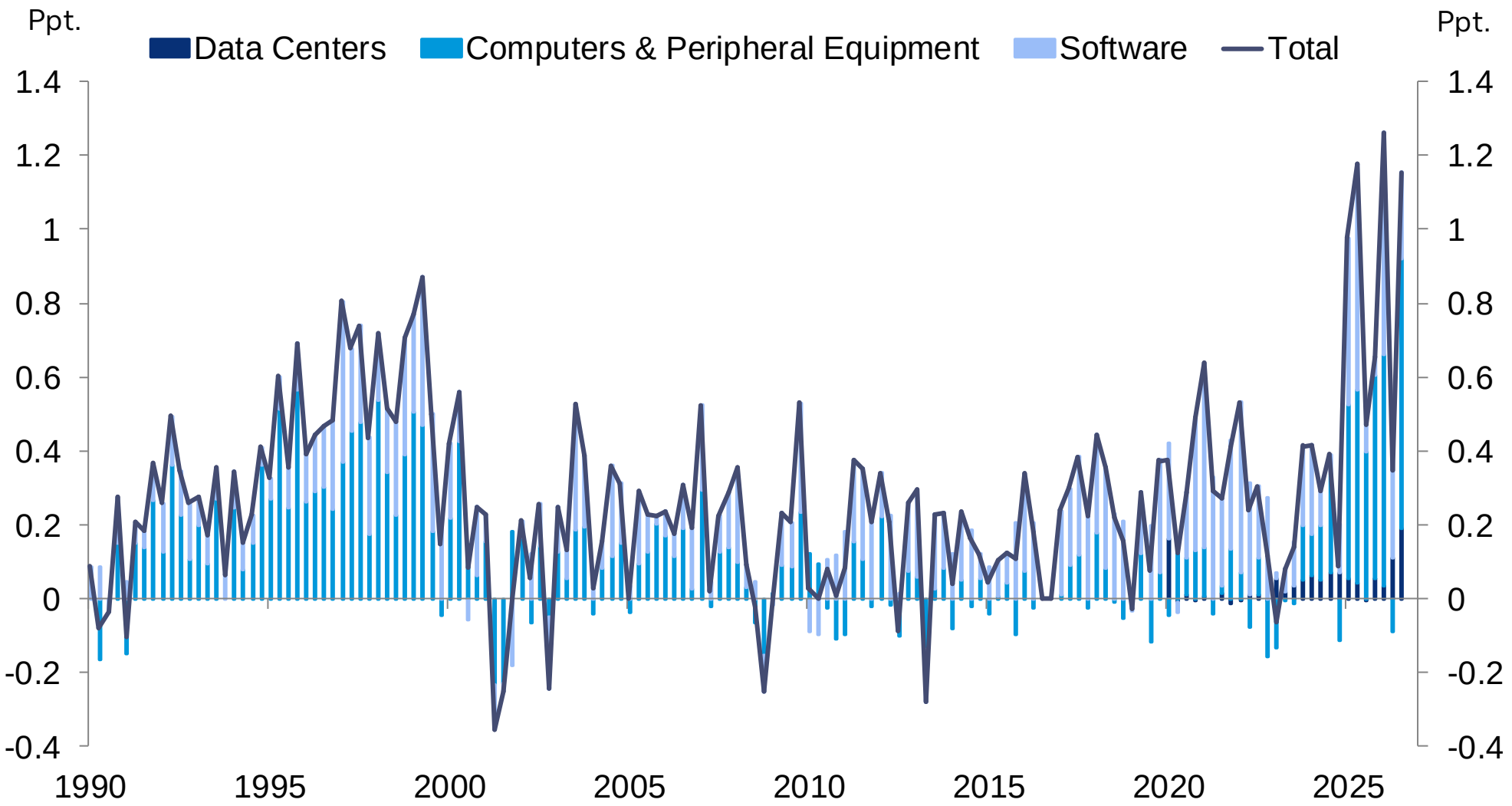


Source: BEA, Haver Analytics, Deutsche Bank

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AI-related capex has been adding meaningfully to growth in recent quarters, though imports reduce the GDP effect

GDP contribution from AI

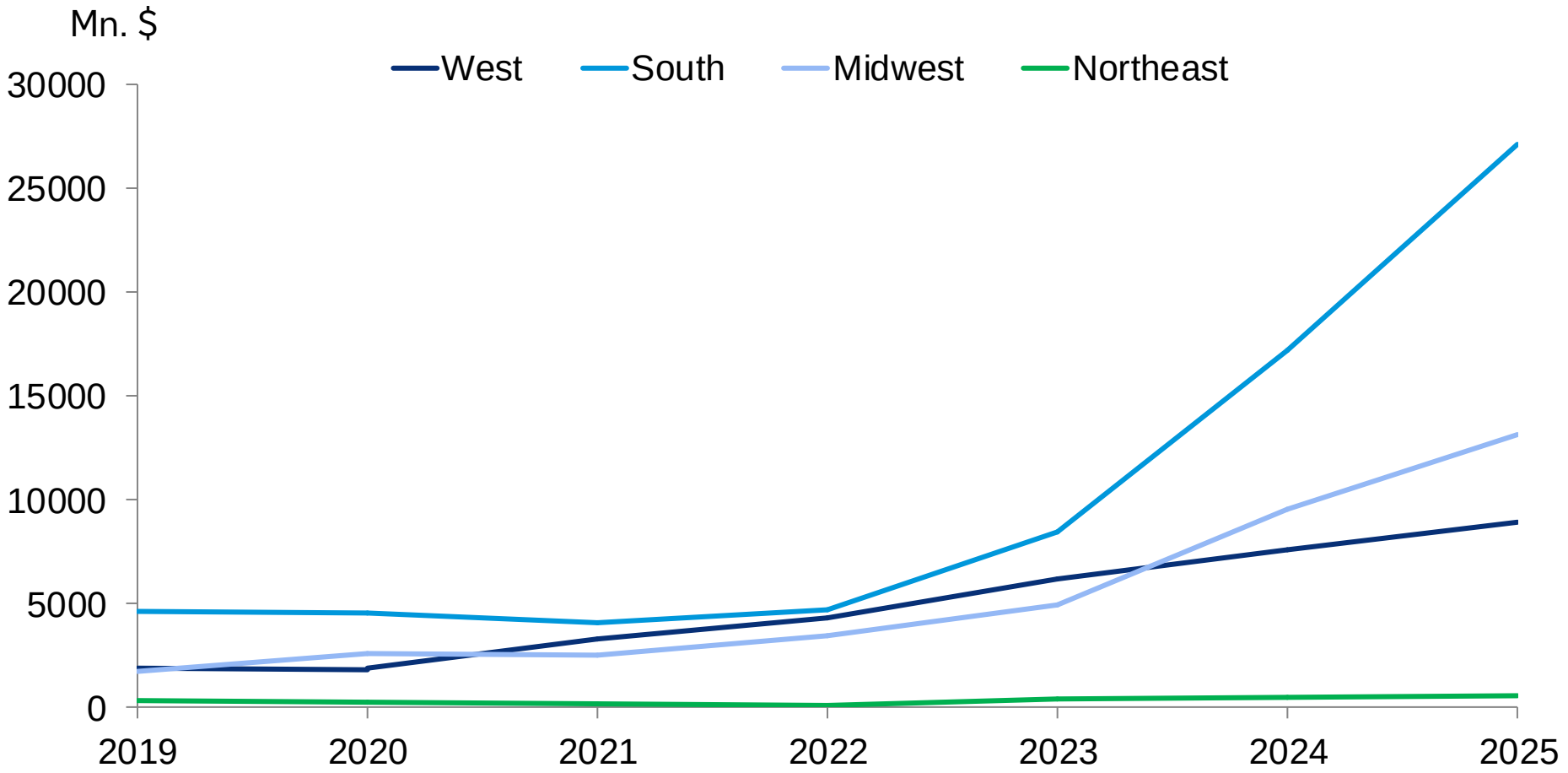


Source: BEA, Haver Analytics, Deutsche Bank

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Data center spending has risen most significantly in South and Midwest

Private nonres. construction put in place: office, data center

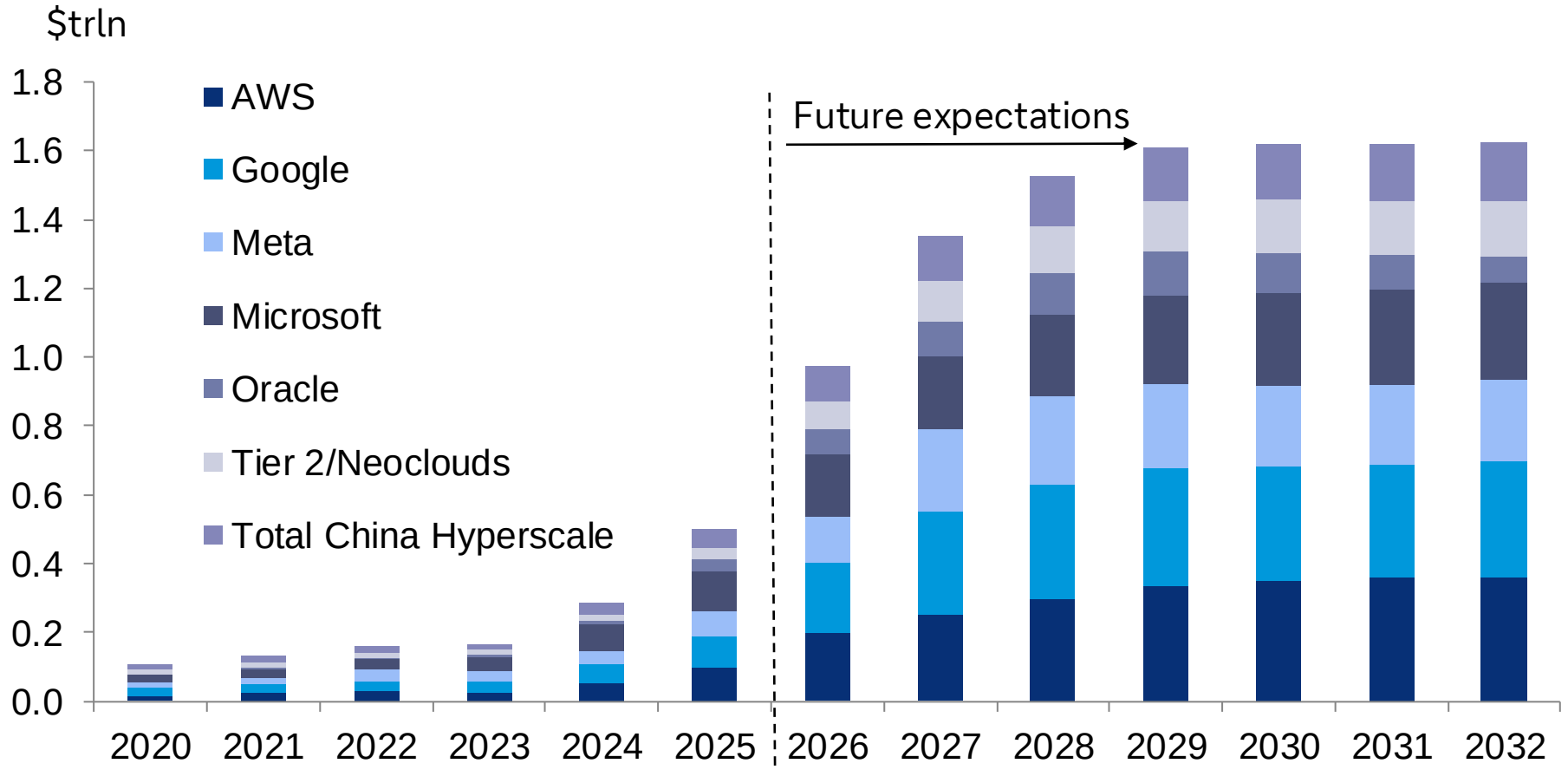


Source: FRB, Census Bureau, Haver Analytics, Deutsche Bank

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Hyperscaler capex levels increase through 2029

Hyperscaler capital expenditure 2020 to 2032 (USD trn)

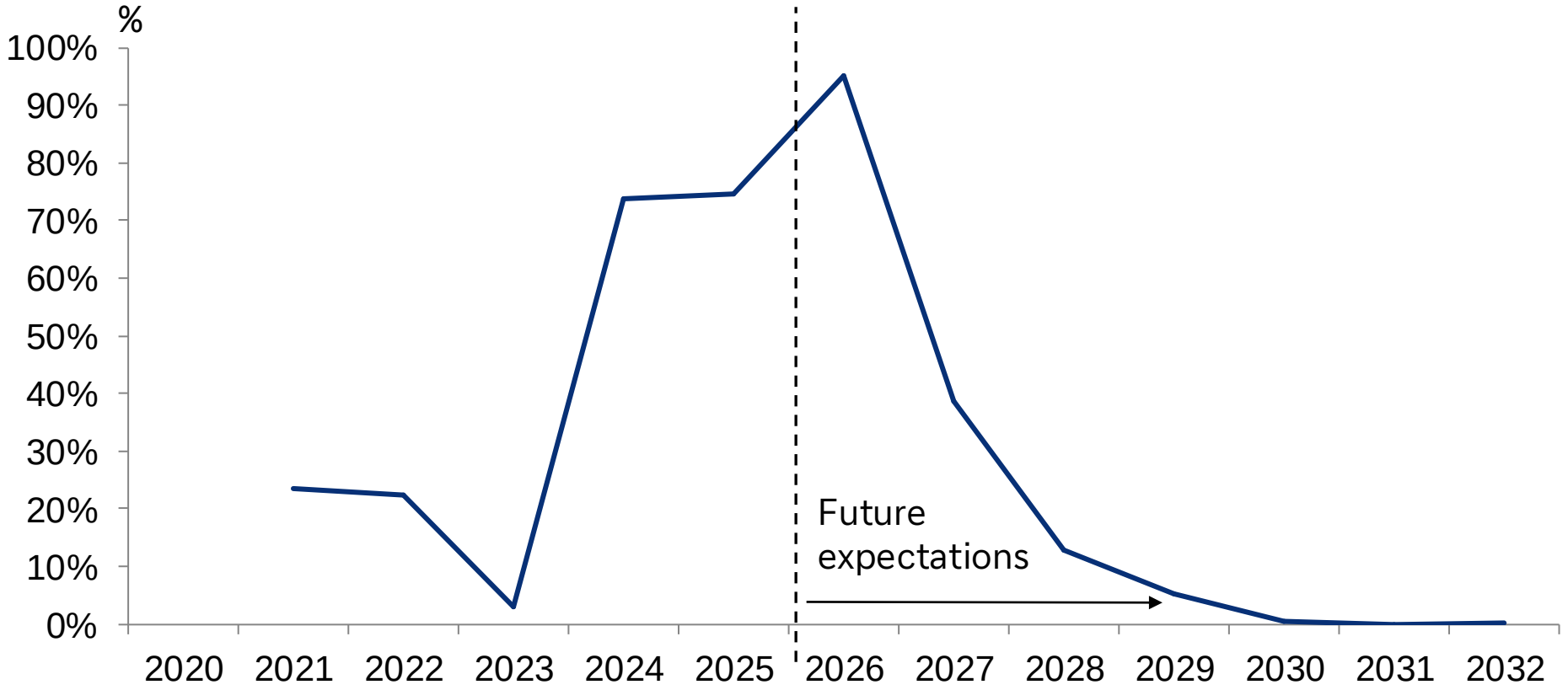


Source: Bloomberg Finance LP, Deutsche Bank

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Hyperscaler capex growth rates slow beginning in 2027

Growth rate in hyperscaler capital expenditure 2020 to 2032

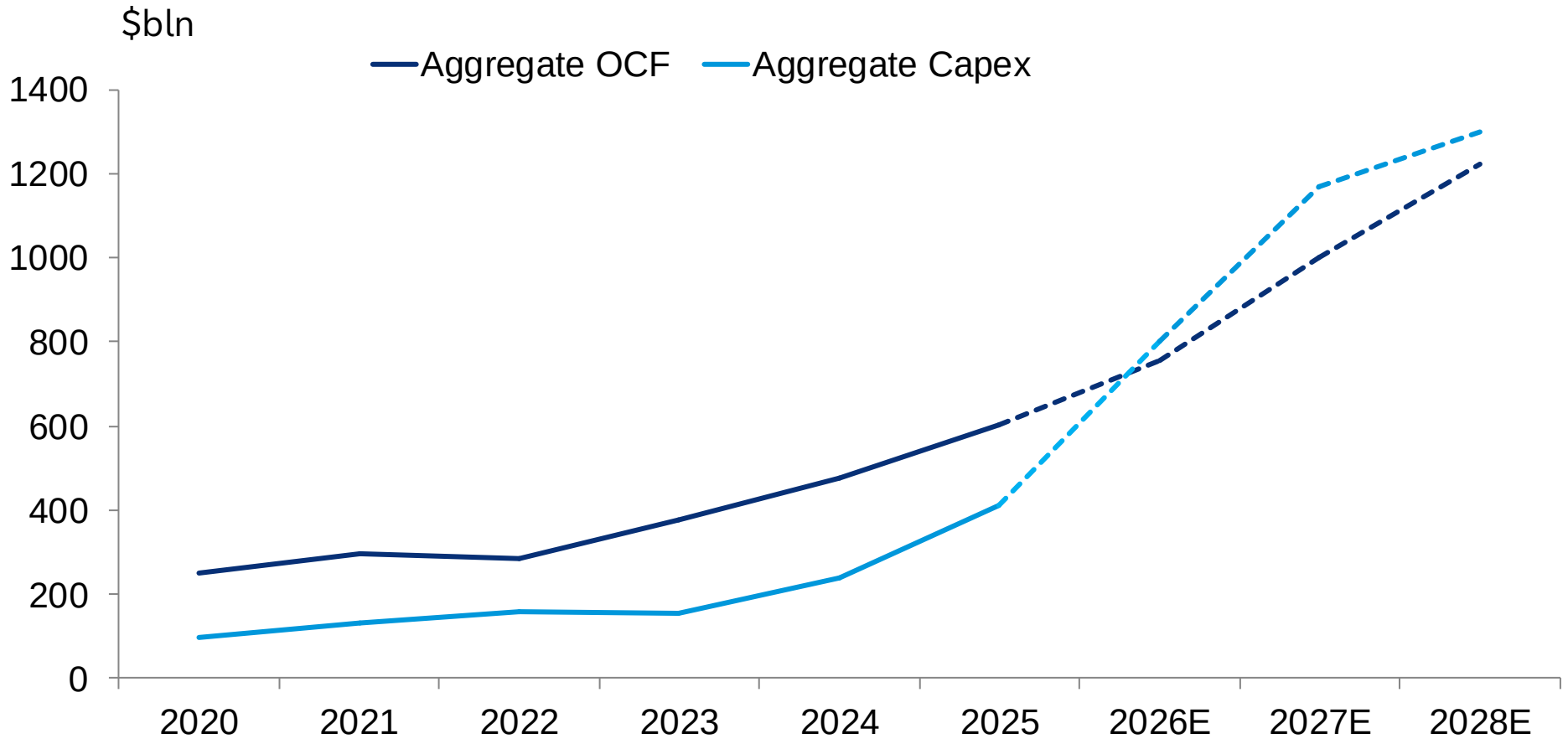


Source: Bloomberg Finance LP, Deutsche Bank

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Free cash flow turns negative as capex ramps up for hyperscalers

Hyperscaler aggregate operating cash flow vs capex (2020-2028E) (USDbn)



Source: Bloomberg Finance LP, Deutsche Bank

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Brookings paper suggests AI capex buildout larger than historical technology capex booms

Selected U.S. Capital-Expenditure Booms

Episode	Period	Size (% of GDP)
Canals	1836–1841	0.66
Railroads	1870–1890	2.24
Electrification	1905–1925	0.50
Highways	1956–1973	1.13
Telecom and fiber	1996–2003	1.10
AI buildout	2025–2032	3.63

Notes: Size is average annual capital expenditure as a percentage of GDP over the indicated period.

Source: Brookings (Van Nieuwerburgh, Stijn. 2026. “Financing the AI Buildout”), Deutsche Bank

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Appendix 1
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